Data Center Operational Blueprint

- User Workflow, User Experience, and Backend Implementation

Deployment Plan User Workflows (Background Information)

User workflow – Manual

BACKGROUND INFORMATION

1. View the cluster manifest
2. View whitespaces – select bundles and capacity cells
   - Review the rack capacity, cell power capacity, cooling capacity, redundancy, and cooling system for the capacity cell.
   - Consider the expansion options: will the tenant be able to expand to bundles
   - Create and start to build the placement plan: pick a capacity cell **or bundle** that may accommodate the cluster manifest (add/ remove CC/rack/bundle to a placement plan).
   - System performs some basic validation check if cc/bundle does not meet requirements. **System computes value and optionality cost**
3. Place Load
   - If fewer racks, manually enter; If more racks, download a template and upload csv.
   - Rack capacity check – load higher than rack capacity not allowed
   - **System computes value and optionality cost**
4. View resources and constraints
   - Use node data panel to see contained and upstream constraints
   - **Use dependency mapping on the spatial view to discover relationships.** Use topology to see relationships. **Focus on highlighting capacity and margins**
5. Adjust placement
   - Manually change kW numbers
   - Drag and drop rack loads – drag and drop rows and zones
   - **Recompute/ update margins and costs**
6. Save draft, save as another draft, save as plan.
7. Export csv of draft and plan

BACKGROUND INFORMATION

The two levels of Placement/ Auto-placement

- There are two levels of decisions during load planning/ placement:
  1) **Macro-placement:** which capacity cell(s)/ bundles to use;
  2) **Micro-placement:** within the CC/bundle, how to distribute racks – which racks where
- Auto-placement also has these two layers.
  1) **Full auto-placement:** the algorithm determines both macro and micro placement
  2) **Partial auto-placement:** the user chooses the CC/bundle (macro-placement). The user may even specify the zones and rows. The algorithm places the racks optimally in the selected areas (micro-placement).

BACKGROUND INFORMATION

User workflow: partial auto-placement

1. View the cluster manifest
2. View whitespaces – select bundles and capacity cells
   - Review the rack capacity, cell power capacity, cooling capacity, redundancy, and cooling system for the capacity cell.
   - Consider the expansion options: will the tenant be able to expand to bundles
   - Create a placement plan
3. Auto-placement configuration
   - Launch auto-placement engine
   - Choose/ confirm capacity cells/ bundles**; Specify zone/row** if desirable. System performs some basic validation check if cc/bundle/zones do not meet requirements. **System computes value and optionality cost.**
   - Choose optimization rule
4. View results
   - Use node data panel to see contained and upstream constraints
   - Use dependency mapping on the spatial view to discover relationships. Use topology to see relationships. Focus on highlighting capacity and margins
   - **View value and optionality costs**
5. Manually adjust placement
   - Manually change kW numbers
   - Drag and drop rack loads – drag and drop rows and zones
   - Recompute/ update margins and costs
6. Save draft, save as another draft, save as plan.
7. Export csv of draft and plan

BACKGROUND INFORMATION

User workflow: full auto-placement

1. View the cluster manifest
2. Auto-placement configuration
   - Launch auto-placement engine
   - Specify if any cc/ bundles/ racks shall NOT be used.
   - Choose optimization rule
3. **Auto-placement engine finds the optimal placement plan per configuration**
4. View results
   - Use node data panel to see contained and upstream constraints
   - **View value and optionality costs**
5. Manually adjust placement
   - Manually change kW numbers
   - Drag and drop rack loads – drag and drop rows and zones
   - Recompute/ update margins and costs
6. Save draft, save as another draft, save as plan.
7. Export csv of draft and plan

UX/UI

UX/UI Items

	Task Priority	Feature importance	Applicable to both Planning and Navigator?
Focus and dimming (not scene switching)	H	H	No
Persistent Labels	M	M	Yes
Expanding Labels to data badges	L	M	Yes
Actions from badges	L	M	Yes
Dependency tracer	L	H	Yes
Completion of data badge content	L	M	Yes, but different data
Load drag and drop	L	M	No

“Focus” (Frame Selection) and Dimming – Clarify Requirements

Effects of using Spatial Topology to navigate: **scene switching** (same as in Navigate: clicking on the nodes changes the scene.)

Effects of using Whitespace and Deployment Blueprints to navigate:

(A) **Focus – “Frame Selection”:** move the camera and adjust distance so the selected scope (as defined below in the table) fits nicely in view.

(B) **Selective dimming: dim the nodes not in the frame selection scope**

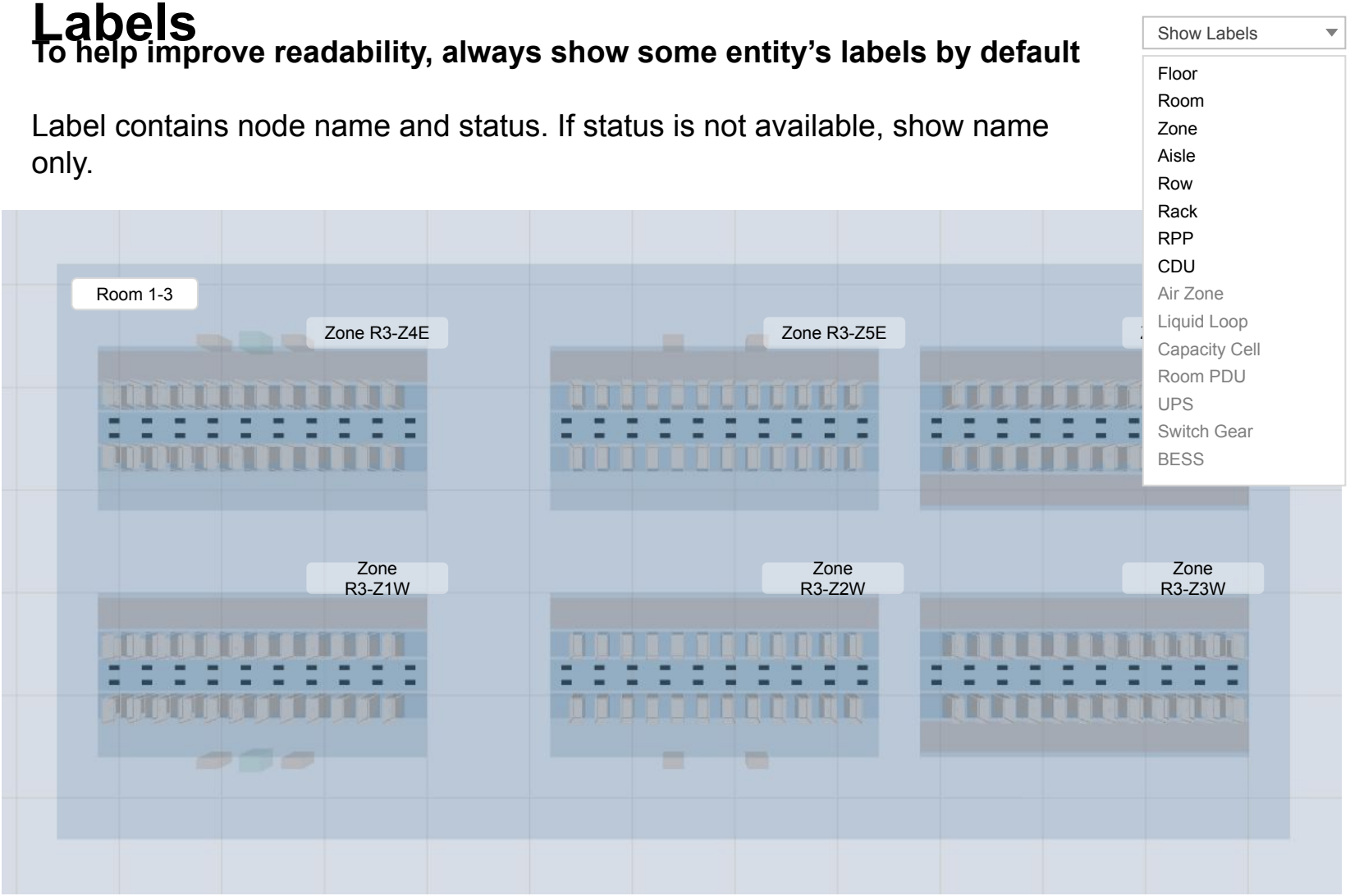
(C) **do NOT switching scenes, instead,** reset the scene to Site (meaning all the nodes should be in accessible if needed)

Selected Node Type in Whitespace and Deployment Navigations	**Frame Selection Scope**
Placement Plan	Entire site
Capacity Cell	All Rooms associated with the Capacity Cell and the substructure of these Rooms in the spatial topology, as well as SANs
Rack	The Row that contains the Rack
Containment Bundles	Entire site
Room Bundle	All Rooms associated with the Bundle and the substructure of these Rooms in the spatial topology, as well as SANs
Room PDU Bundle	All Rooms associated with the Bundle and the substructure of these Rooms in the spatial topology, as well as SANs
USP Bundle	All Rooms associated with the Bundle and the substructure of these Rooms in the spatial topology, as well as SANs

Persistent Labels

To help improve readability, always show some entity's labels by default

Label contains node name and status. If status is not available, show name only.



DEFAULT

(1) Show labels of
- Room
- Zone

(2) When these SANs are selected, also show their labels by default:
- Capacity Cell
- Air Zone
- Liquid Loop

(3) When a SAN is not selected, there is no option to show the label

Provide an option (drop down) for the user to show/hide labels by type.

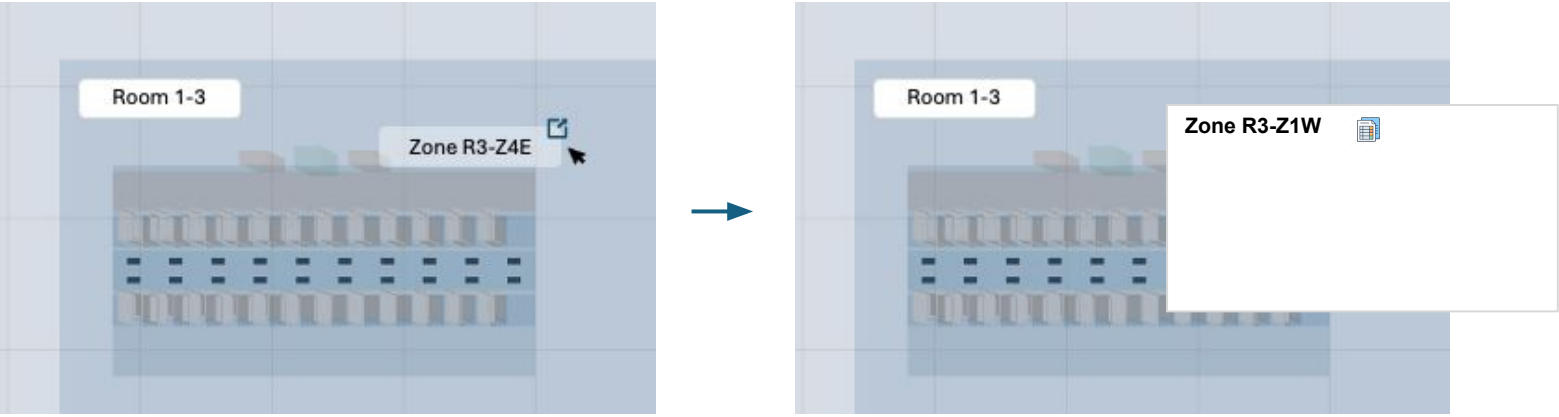
Labels not "shown" will still appear upon cursor over.

Expanding Labels to data badges

For each node, there are two types of annotated information: **labels and badges.**

- The label should contain only name and state. If state is not available, show name only.
- The badge should contain data. In the Planning view, show capacity and associated infrastructure data.
- **Default:** some node types by default show labels. Other node types by default show badges. Store this default setting as node data.

- When a label is visible, the user cursor over reveals the **expand icon**. Click on the icon to expand the label into a badge. This expansion is temporary to the interaction. The next time the node annotation behavior will go back to the default (i.e. a label).
- Clicking directly ON the label itself will open the node data panel.



Node Types with Default Data Badges in the Planning Section

- Rack
- Capacity Cell
- RPP
- Room PDU
- Air Zone
- Liquid Loop

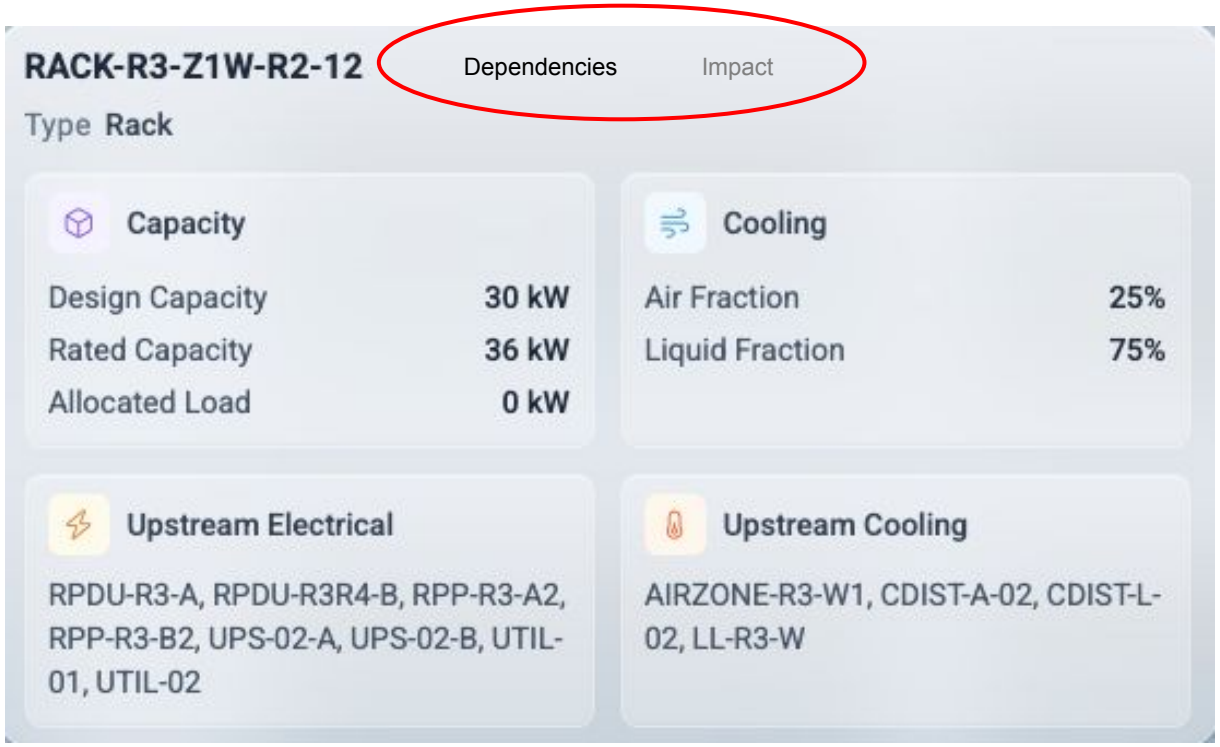
Other node types will show labels by default.

Currently we are not supporting an end-user option to change this default behavior, so the default will be determined at setup time as a configuration.

Actions from Badges

- Clicking on the badge itself opens the node data panel.

- Add two special actions: Trace Dependencies and Trace Impacts. These are functions of the Dependency Tracer



Dependency Tracer (WIP)

Dependency Tracer allows the users to highlight the **dependencies and impacts** of a selected node.

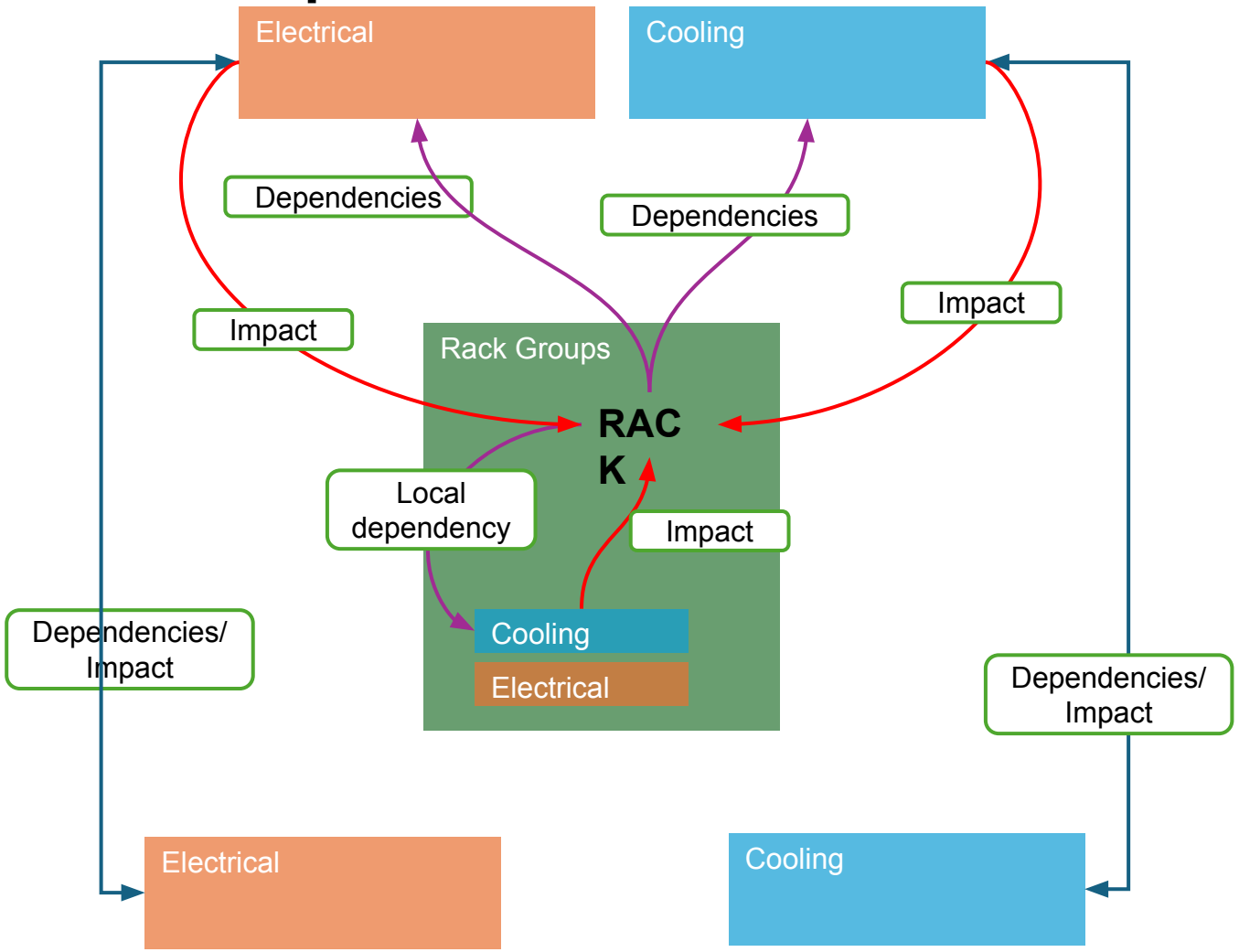
Example:

The user selected a rack, and clicked on "**Dependencies**".

- The Air zone and LL are highlighted (blue)
- The CDU and Air Cooling Units are also highlighted, if they are within view (blue)
- The RPP and Room PDU are highlighted (Red)

- **The trace should work in 3D view including the top view.**

Dependency Tracer Concepts



Rack Groups: entities that can be seen as a collection of racks

- Row
- Zone
- Capacity Cell
- Room
- Room Bundle
- Room PDU Bundle
- UPS Bundle

Impact and Dependency – two sides of the same coin

Impact

- Most nodes in the electrical and cooling systems can **impact** racks and rack groups Load Impact
- Most nodes in the electrical and cooling systems can also **impact** their own downstream nodes Downstream Impact

Dependency

- Rack and rack groups have **dependencies** on electrical and cooling system node. Some are Local Dependencies; some are Upstream Dependencies.
- Electrical and cooling nodes also have **dependencies** on their own upstream nodes Upstream Dependencies

Dependency Rules

For dependencies maps of the current topologies, see google sheet: https://docs.google.com/spreadsheets/d/1-gKV23TcmoDeI9TCRfp6uii8TWyNC3tg9SH-E6Cnb0A/edit?gid=835908655#gid=835908655

Relevant Node Type	Dependency - Upstream	Dependency - Local
Rack	All electrical nodes upstream to the Rack PDUs of the Rack All cooling nodes upstream to the Rack	Rack PDU Cooling nodes downstream to the Rack
Row, Zone	Same as above	Same as above
Capacity Cell, Room Bundle	All electrical nodes upstream to RPP All cooling nodes upstream to Air Zone and LL	RPP Air Zone, LL
Room PDU Bundle, UPS Bundle	All electrical nodes upstream to UPS All cooling nodes upstream to Air Zone and LL	RPP, UPS Air Zone, LL
Electrical Nodes from Rack PDU and up	All electrical nodes upstream to the selected node	N/A
Cooling Nodes from RDHx/ DTC and up	All cooling nodes upstream to the selected node	N/A

Impact Rules

For impact maps of the current topologies, see google sheet: https://docs.google.com/spreadsheets/d/1-gKV23TcmoDeI9TCRfp6uii8TWyNC3tg9SH-E6Cnb0A/edit?gid=835908655#gid=835908655

Relevant Node Type	Impact - Load	Impact - Downstream
Electrical Nodes from Rack PDU and up which are Capacity Nodes	Rack Rack Groups	Electrical nodes downstream to the selected node
Electrical Nodes from Rack PDU and up which are NOT Capacity Nodes	N/A	Electrical nodes downstream to the selected node
Cooling Nodes from RDHx/ DTC and up which are Capacity Nodes	Rack Rack Groups	Cooling nodes downstream to the selected node
Cooling Nodes from RDHx/ DTC and up which are NOT Capacity Nodes	N/A	Cooling nodes downstream to the selected node

Dependency and Impact maps should be created as system data and remain updated if the topologies change

Capacity Nodes and Active Constraints

Capacity domains (or Capacity nodes) are nodes where you track *supply and load balance*. They represent a domain or boundary where you can say:
“X kW available, Y kW consumed.”

Some Capacity nodes are Active Constraints, meaning they themselves limit the capacity of others.
Other Capacity nodes are Passive Constraints, meaning they receive constraints being imposed by other nodes.

Node Type	Topology	Capacity Node (Capacity Domain)	ActiveConstraint
Rack PDU	Electrical System	FALSE	FALSE
RPP	Electrical System	TRUE	TRUE
Room PDU	Electrical System	TRUE	TRUE
UPS	Electrical System	TRUE	TRUE
BESS	Electrical System	TRUE	TRUE
Switch Gear	Electrical System	FALSE	FALSE
Utility Feed	Electrical System	TRUE	TRUE
Generator	Electrical System	TRUE	TRUE
RDHx	Cooling System	FALSE	FALSE
DTC	Cooling System	FALSE	FALSE
Air Zone	Cooling System	TRUE	TRUE
Liquid Lood	Cooling System	TRUE	TRUE
Air Cooling Unit	Cooling System	FALSE	FALSE
CDU	Cooling System	FALSE	FALSE
Cooling Distribution	Cooling System	TRUE	TRUE
Cooling Plant	Cooling System	TRUE	TRUE
Rack	Spatial Topology	TRUE	FALSE
Row	Spatial Topology	FALSE	FALSE
Zone	Spatial Topology	FALSE	FALSE
Aisle	Spatial Topology	FALSE	FALSE
Capacity Cell	Whitespace Blueprint	TRUE	FALSE
Room Bundle	Whitespace Blueprint	TRUE	FALSE
Room PDU Bundle	Whitespace Blueprint	TRUE	FALSE
UPS Bundle	Whitespace Blueprint	TRUE	FALSE

Capacity Nodes and Active Constraints may change if topologies change.

They will remain a manual configuration of the system data to be updated.

Passive constraints

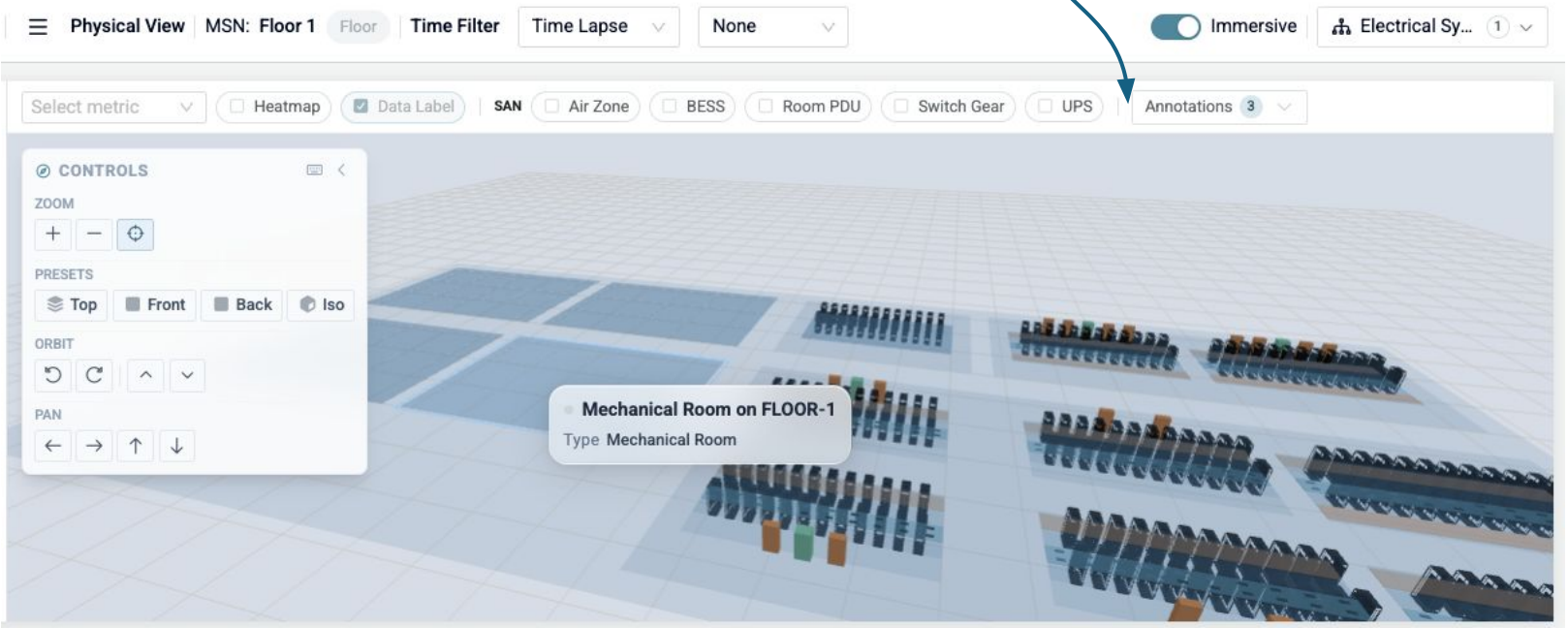
Two Ways to Show Dependencies/ Impacts

1. Interaction Bar

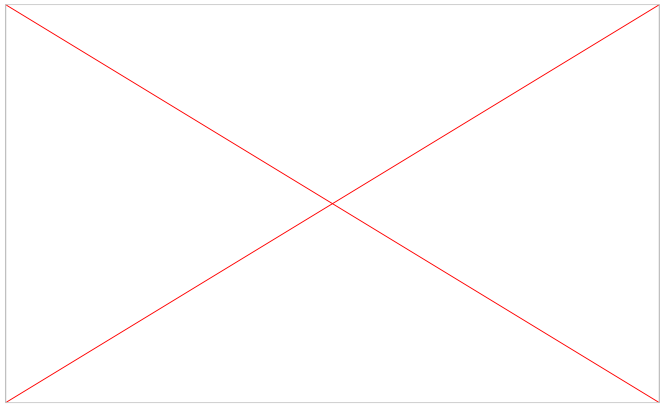
a. The user clicks and selects an entity in the 3D view (called “Source Node” or “Traced Source Node – TSN”) (the source node should be property highlighted)

b. The user clicks on the “Dependency” or ”Impact” button above.

Since the # of interaction buttons are growing, we may need to create a secondary layer, such as using tabs to show different menu items



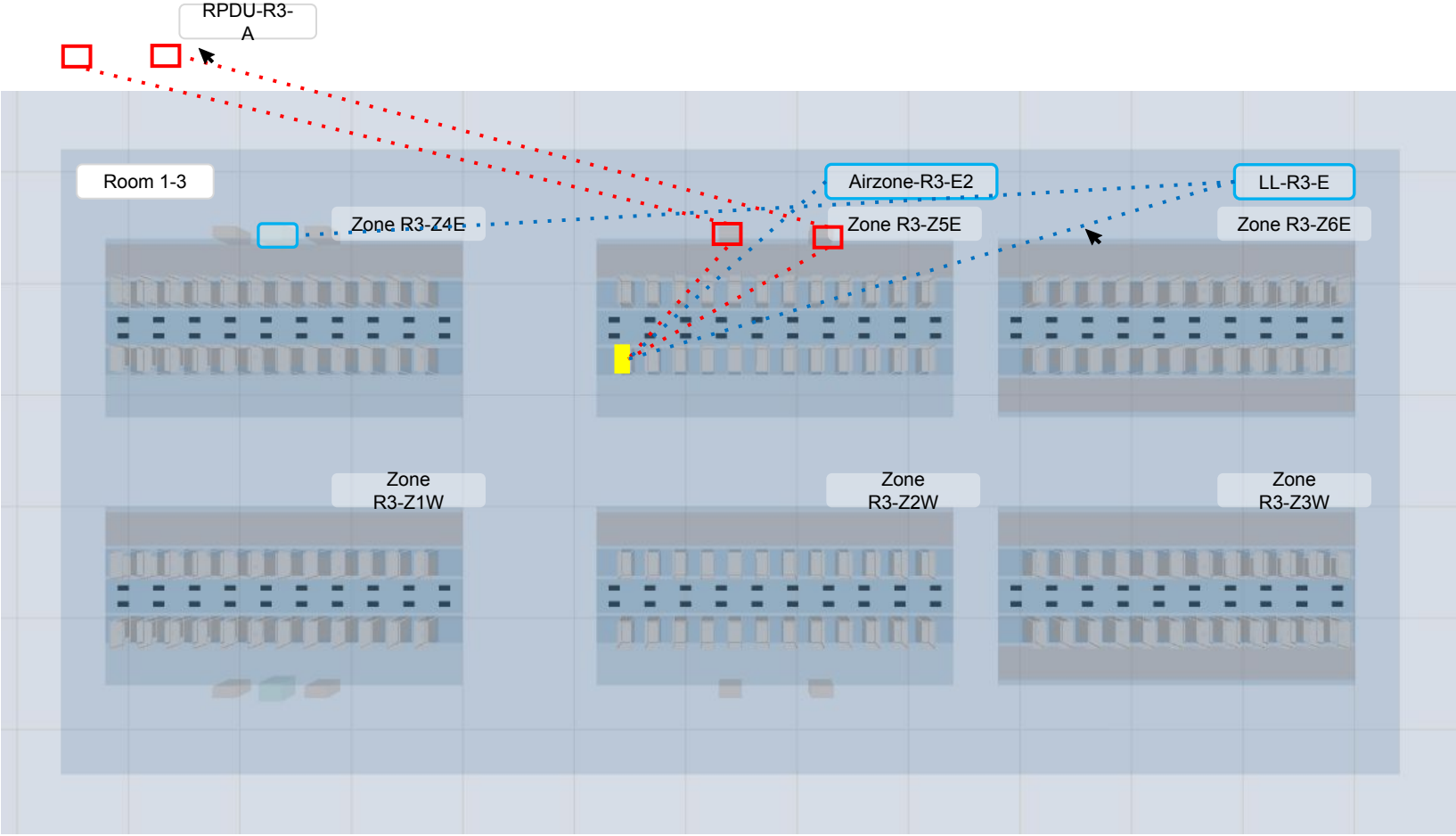
2. From the data badge of the DSN



Tracing Behaviors: Two Parts of of Dependency/ Impact Effects

Part 1 – contour highlight: the dependent or impacted entities have bright contour highlights

- Use different colors for different topology (e.g. red/orange for electrical, blue for cooling)
- For single geo/box entities, show contour around the object itself
- For SANs that are memberships (Air Zone, LL, Capacity Cell, the containment bundles), highlight both the label and the contour of the shape.
- For entities that are areas (row, zone, room), also highlight both the label and the contour of the shape.



Part 2 – tracing lines: lines that connect the entities according to topology (edges)

- Try to render tracing lines as native objects in the 3D view as well, so that they can be visible from different 3D angles.
- Straightlines are OK for now, no need to use curves/joints.
- For SANs that are memberships, and for entities that are areas, **connect the lines to the label, not to the shape**

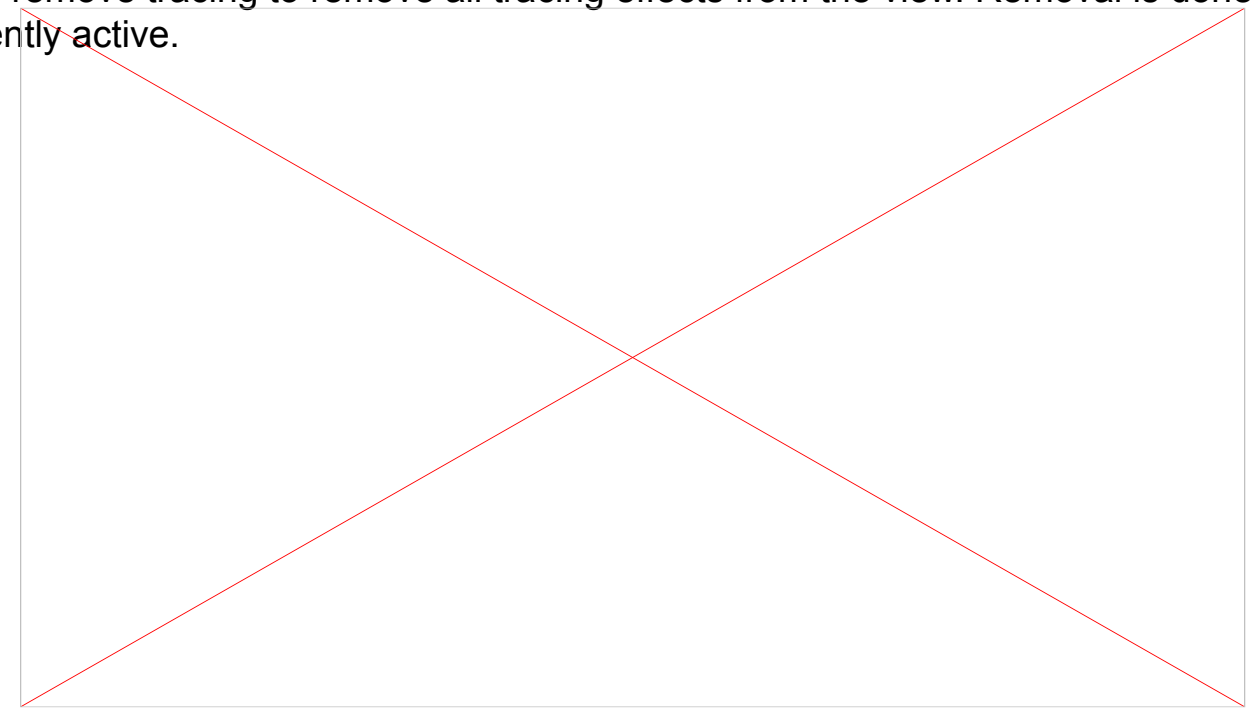
Tracing Behaviors

Tracing Levels

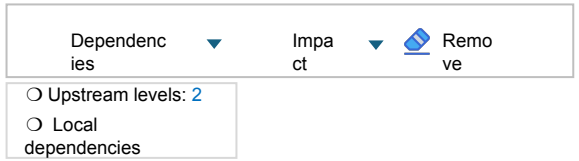
Upstream and downstream node/edges to the source node should carry a Level # in the data, which indicates how many degrees away they are from the Source Node. By default, **show two levels upstream and two levels downstream (**But this setting can be changed by the user)

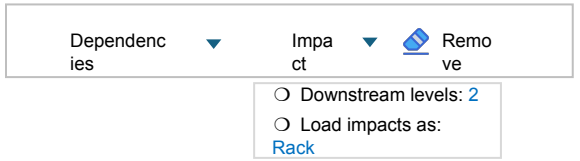
Multiple Tracing: Tracing can be shown for **multiple Source Nodes**. For example, once the upstream of a rack is traced, the user can continue on and select another rack to trace upstream, or they can select one of the upstream nodes (e.g. RPP), and rack its own upstream.

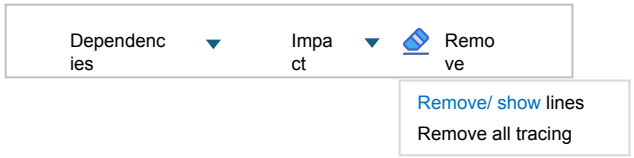
Remove Tracing: Click on remove tracing to remove all tracing effects from the view. Removal is done for dependencies/impacts of ALL source nodes that are currently active.



Tracing Options – these settings applies to **all active TSNs** at the same time







Dependency Options

- Any Traced Source Node (TSN) can have two types of dependencies: Upstream and Local. The user can choose to show both types or only one of them.
- If showing upstream dependencies, the user can also choose how many levels of upstream are shown.
- If local dependencies are checked, all local dependencies will be shown. There is no “level” with local dependencies.

Impact Options

- Any TSN can have two types of impacts: Downstream and Load Impacts. The user can choose to show both types or only one of them.
- If showing downstream impacts, the user can also choose how many levels of downstream are shown.
- “Load Impacts” means the TSN impacts racks and rack groups. If showing Load Impacts, the scope of the impacts has to be selected. Only one option can be chosen?
    - Rack: means the individual impacted racks are highlighted one by one
    - Row or Zone or Room: means to treat a row/ a zone/ a room as a single entity. If any rack within the row/zone/room is impacted, the entire group is considered impacted.
    - Capacity cell or room bundle, or room PDU bundle, or UPS bundle: means to treat the CC or the bundle as a single entity. If any rack within is impacted, the entire group is considered impacted.

- **Remove Options**
    - Remove all tracing means all highlights and lines will be removed
    - Remove lines means only the lines will be hidden, but the highlights will remain
    - Show lines meaning for the existing active TSN tracing only, bring back the lines

Load Drag and Drop

When the user edits rack load, enable drag-and-drop of the rack loads. After drop, provide options to either (1) reverse change or (2) update load (recompute)

Scope to be dragged:
Provide interaction so that the user can select a single rack, a single row, or an entire zone's rack loads at one time.

Places to drop:
A rack load can be dropped onto any rack with no load. A rack load cannot be dropped onto another rack with load.

- Validation: the load must be equal or smaller than the rack's circuit capacity

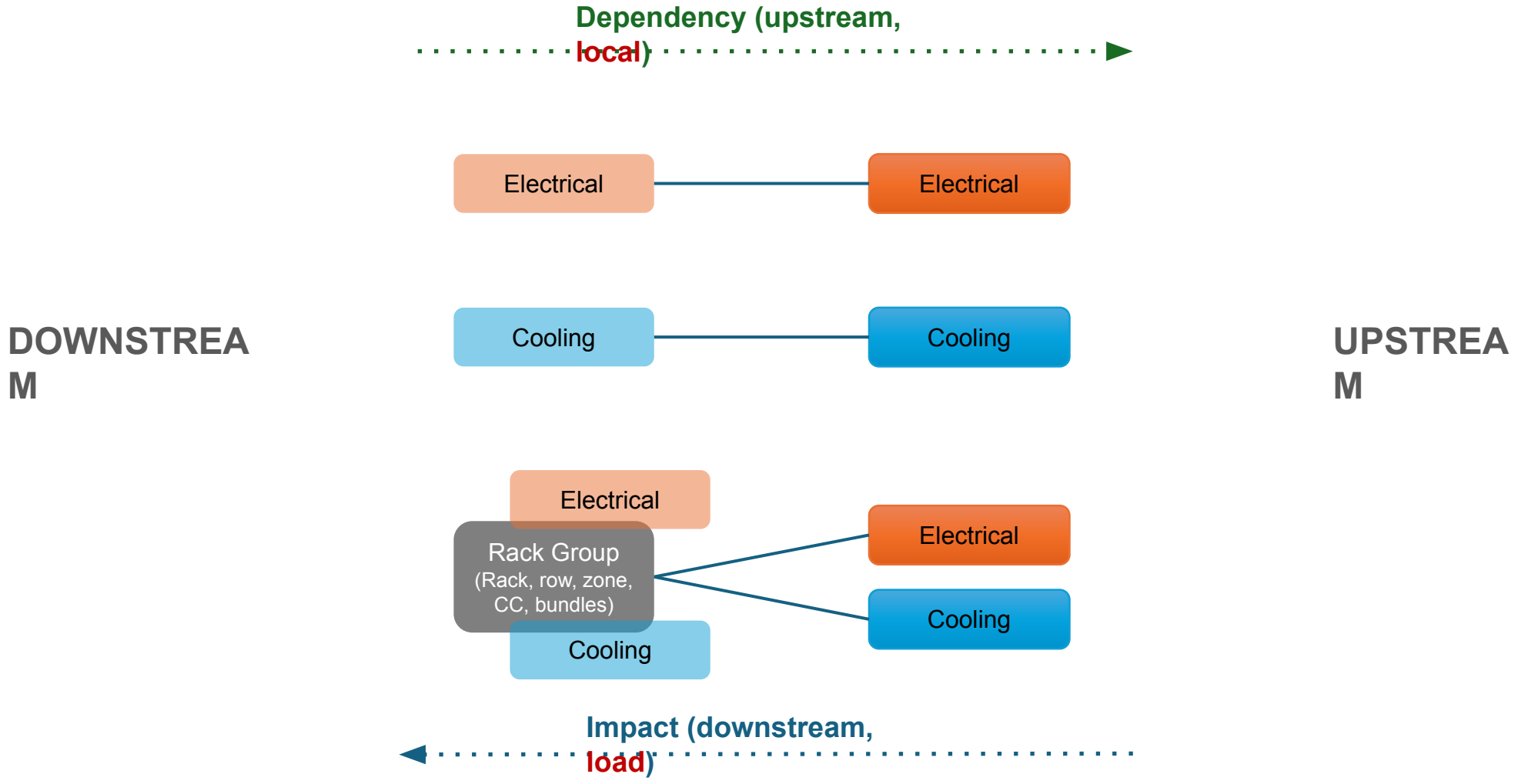
A row load can potentially be dropped onto another row, including partially loaded rows.

- Validation: the rack loads most be able to fit into the available empty racks in the target row.
  1) There has to be enough unloaded/free rack positions available
  2) The rack loads have to be able to fit into the available racks' circuit capacity. If these loads or capacity are not uniform across, then both of them have to be sorted from big to small. Then start by dropping the biggest load to the biggest capacity and work your way down.
  3) The spatial adjacency of the rack loads do not need to be maintained. Find available rack positions and fit in.

A zone can potentially be dropped onto another zone, including partially loaded zones.

- Validation: the rack loads most be able to fit into the available empty racks in the target zone.
  1) There has to be enough unloaded/free rack positions available
  2) The rack loads have to be able to fit into the available racks' circuit capacity. If these loads or capacity are not uniform across, then both of them have to be sorted from big to small. Then start by dropping the biggest load to the biggest capacity and work your way down.
  3) The spatial adjacency of the rack loads do not need to be maintained. Find available rack positions and fit in.

Conceptual Clarifications of Dependencies and Impacts



Backend (WIP)

Backend Items/ Modules

1. **Load-Capacity Calculator**
   - Uses the appropriate input data to compute capacity and load across nodes.
   - Some computations are (a) topology consolidation; Other computations involve (b) domain-specific logic
   - Create necessary triggers for the data updates
   - Validations for load placement
2. Variable data, data models
   - Clean up/ consolidate/ complete the variables
   - Other data models needed for backend
3. OP and Metrics
   - Clarify the technical difference between OP and Metrics and make changes if needed
4. Optionality value calculator OAKw Evaluator
   - Compute optionality-adjusted kW (OAKw) value for any subdomain of the data center based on optionality rules.

Backend Items/ Modules

5. Auto-placement engine, module 1: Optionality Value Optimizer
6. Auto-placement engine, module 2: Capacity Packer
7. Blueprint Compiler Whitespace Packing
   - Given infrastructure input parameters, automatically organizes the data center into capacity cells and containment bundles
8. Topology update manager – if needed for future updates to be more scalable

Load Capacity Calculator has the following steps

1. **Capacity**
   1) Roll up Rack_Circuit_Capacity_design from Rack
   2) Compute/ roll up Rack Positions
   3) Compute Capacity Cell Power_Capacity, Thermal_Capacity, and Capacity Envelope

2. **PD/PP Load** Computations: these are PD/PP specific – for each PD/PP
   1) Compute Allocated_AirCool_Load and Allocated_LiquidCool_Load for Racks
   2) Roll up Allocated_ITLoad, Allocated_AirCool_Load and Allocated_LiquidCool_Load from rack
   3) Roll up Rack_Count

3. **Consolidated Load**
   1) Consolidate Allocated_ITLoad, Allocated_AirCool_Load, Allocated_LiquidCool_Load and Rack_Count cross PP's
   2) Consolidate Allocated_ITLoad, Allocated_AirCool_Load, Allocated_LiquidCool_Load and Rack_Count for each PD

4. **Margins**
   1) Compute Local Capacity Margins
   2) Compute Device Capacity Margins
   3) Compute Cell Margins, Power Margins, Thermal Margins

5. **Upstream and Subdomain Margins**
   1) Prepare Upstream Capacity, Load and Margins
   2) Prepare Subdomain Capacity, Load and Margins

BACKGROUND INFORMATION

Variables Used in Load Capacity Calculator

In google sheet:
https://docs.google.com/spreadsheets/d/1-gKV23TcmoDeI9TCRfp6uii8TWyNC3tg9SH-E6Cnb0A/edit?gid=1236255353#gid=1236255353
Tab [Variables].

The Variables have been marked as Input, Reference, or Metric.

- **Reference:** constant values set at the beginning of the use
- **Input:** user-entered value or externally ingested value during the user interactions
- **Metrics:** values calculated using Reference or Input

The Load Capacity Calculator will assume the Reference and Input are already present, and use them to calculate relevant metrics.

1. Capacity

1) **Roll up Rack_Circuit_Capacity_(design) from Rack**
2) **Compute/ roll up Rack Positions**
   - Each rack count as rack position = 1

Consolidation Variable	Consolidation Context	Node Types in Scope
Rack_Cirtuit_Capacity_(design)	Rack rollup – Spatial	Rack, Row, Zone, room, Floor, IT Facility, Site
	Rack rollup - Whitespace	Rack, Capacity Cell, Room Bundle, Room PDU Bundle, UPS Bundle, Containment Bundles
Rack Positions	Rack rollup – Spatial	Rack, Row, Zone, room, Floor, IT Facility, Site
	Rack rollup – Whitespace	Rack, Capacity Cell, Room Bundle, Room PDU Bundle, UPS Bundle, Containment Bundles

For **consolidation context**, refer to:
https://docs.google.com/spreadsheets/d/1-gKV23TcmoDeI9TCRfp6uii8TWyNC3tg9SH-E6Cnb0A/edit?gid=8287137#gid=8287137
Tab [ContextRule]

1. Capacity

3) Compute Capacity Cell Power_Capacity, Thermal_Capacity, and Capacity Envelope

Capacity_Cell.Power_Capacity = $\sum(RPP_{Panel_{Capacity_{design}}})$

Capacity_Cell.Thermal_Capacity = $\sum(AirZone_{Cooling}Capacity) + \sum(LL_{Cooling_{Capacity}})$

Capacity_Cell.Capacity_Envelope = MIN (Power_Capacity, Thermal_Capacity)

2. PP/PD Load Computations

1) **Compute Allocated_AirCool_Load and Allocated_LiquidCool_Load for Racks**

Allocated_AirCool_Load = Allocated_ITLoad x Rack_AirCool_Fraction

Allocated_LiquidCool_Load = Allocated_ITLoad x Rack_LiquidCool_Fraction

For each Rack, there needs to be multiple ”versions” of Allocated_ITLoad, Allocated_AirCool_Load and Allocated_LiquidCool_Load.

Placement Plan version: this one version should have allocated load from ALL and Any Placement Plans. (One rack cannot be used by multiple placement plans)

Placement Draft versions: EACH placement draft shall have its own rack load versions. For example, the same rack may have 30 kw allocated in PD01, but 0 kw allocated in PD02. Such loads are “private” to the specific Placement Draft.

Any upstream nodes that acquire rollup values of these rack loads will also acquire multiple versions.

For **consolidation context**, refer to:
https://docs.google.com/spreadsheets/d/1-gKV23TcmoDeI9TCRfp6uii8TWyNC3tg9SH-E6Cn/edit?gid=8287137#gid=8287137
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2. PP/PD Load Computations

2) **Roll up Allocated_ITLoad, Allocated_AirCool_Load and Allocated_LiquidCool_Load from Rack**
3) **Roll up Rack_Count** (any rack with allocated load will count as "1")

Consolidation Variable	**Consolidation Context**	**Node Types in Scope**
PP.Allocated_ITLoad PP.Allocated_AirCool_Load PP.Allocated_LiquidCool_Load Rack_Count	Rack rollup – Spatial	Rack, Row, Zone, room, Floor, IT Facility, Site
PP.Allocated_ITLoad PP.Allocated_AirCool_Load PP.Allocated_LiquidCool_Load Rack_Count	Rack rollup - Whitespace	Rack, Capacity Cell, Room Bundle, Room PDU Bundle, UPS Bundle, Containment Bundles
PP.Allocated_ITLoad PP.Allocated_AirCool_Load PP.Allocated_LiquidCool_Load Rack_Count	Rack rollup - Placement Plan	Deployment, Cluster, Placement Plan, Capacity Cell, Rack
PD.Allocated_ITLoad PD.Allocated_AirCool_Load PD.Allocated_LiquidCool_Load Rack_Count	Rack rollup - Placement Draft	Placement Draft, Capacity Cell, Rack
PP.Allocated_ITLoad PD.Allocated_ITLoad Rack_Count	Rack rollup - Electrical	Rack, Rack PDU, RPP, Room PDU, UPS
PP.Allocated_AirCool_Load PD.Allocated_AirCool_Load Rack_Count	Rack rollup – Air Cooling	Rack, Air Zone, Air Cooling Unit, Cooling Distribution, Cooling Plant
PP.Allocated_LiquidCool_Load PD.Allocated_LiquidCool_Load Rack_Count	Rack rollup – Liquid Cooling	Rack, Liquid Loop, CDU, Cooling Distribution, Cooling Plant

3. Consolidated Load

1) Planned Load: Consolidate Allocated_ITLoad, Allocated_AirCool_Load, Allocated_LiquidCool_Load and Rack_Count cross all **Placement Plans**

Consolidated Planned Load = $\sum load\ of\ all\ Clusters\ and\ their\ \boldsymbol{Placement\ Plans}$

For example, the RPP's consolidated Allocated ITLoad is the total of Cluster 1 PP01 and Cluster 1 PP02.
But it does not include any load from Cluster 1 PD01 or Cluster 2 PD01

Planned Load is a "shared" or "public" reality for everyone.

3. Consolidated Load

2) Potential Load: Consolidate Allocated_ITLoad, Allocated_AirCool_Load, Allocated_LiquidCool_Load and Rack_Count **assuming each Placement Draft**

Consolidated Potential Load $_{PD(x)}$ = $Consolidated\ planned\ load\ + \sum load\ of\ PD(x)$

For example, the RPP's consolidated Potential ITLoad with regard to Cluster 3 PD1 is the consolidated planned load calculate in (1) ***Plus*** *the drafted load planned in PD1*

Potential Loads are always specific to a particular Placement Draft. It is "private" to the placement draft.

4. Local/ Capacity Margins

1) Compute Local Capacity Margins
2) Compute Device Capacity Margins
3) Compute Cell Margins, Power Margins, Thermal Margins

For Formula, refer to excel: Node data panel capacity data_v2.2, Tab [CM]

Upstream Margins

A. Create **Upstream Type reference table (**this table should be updated when topology scheme changes)

An example of this table is created in the excel Capacity simulation_topology v2.2, Tab [UpstreamType]. However, we need a function to create & update this table because topologies may change from customer to customer.

Upstream Type Rule

a node **may** have its upstream capacity and capacity margin if

a. it is a Capacity Node itself AND
b. it has an upstream Electrical or Cooling upstream node, either through its own topology or through ”portaling” into the electrical or cooling system topologies, AND
c. The Electrical or Cooling node is also an Active Constraint

B. Use the Upstream Type reference table to find the upstream nodes and their capacity, capacity margins

Subdomain Margins

A. Create **Subdomain Type reference table (**this table should be updated when topology scheme changes)

An example of this table is created in the excel Capacity simulation_topology v2.2, Tab [UpstreamType]. However, we need a function to create & update this table because topologies may change from customer to customer.

Subdomain Type Rule

A node **may** have its subdomain capacity and capacity margin if

It is in the Whitespace Blueprint topology and it is a descendant of the node type “Containment Bundles”

(currently this would include Room Bundle, Room PDU Bundle, UPS Bundle. Capacity Cell, and Rack)

The subdomain nodes of this node should include: its own descendants within the Whitespace Blueprint that are also **Capacity Nodes**

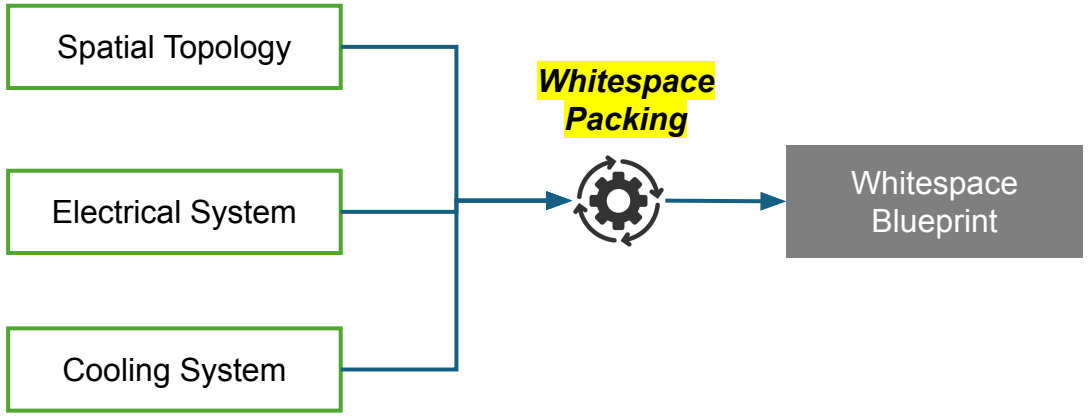
B. Use the Subdomain Type reference table to find the subdomain nodes and their capacity, capacity margins

Whitespace Packing

Introduction

Most topology instance data (including spatial, electrical, cooling instances) are created using direct customer data.

Whitespace blueprint instance data are derived from other topologies, using Whitespace Packing modules.



Whitespace Packing “create” capacities cells (CCs), containment bundles (CBDs) and their composites, based business rules and on the relationships in other topologies.

Supply Block Options

Base Supply Block Options: All CCs and CBDs are Base Supply Block Options (**Base SBOs**).

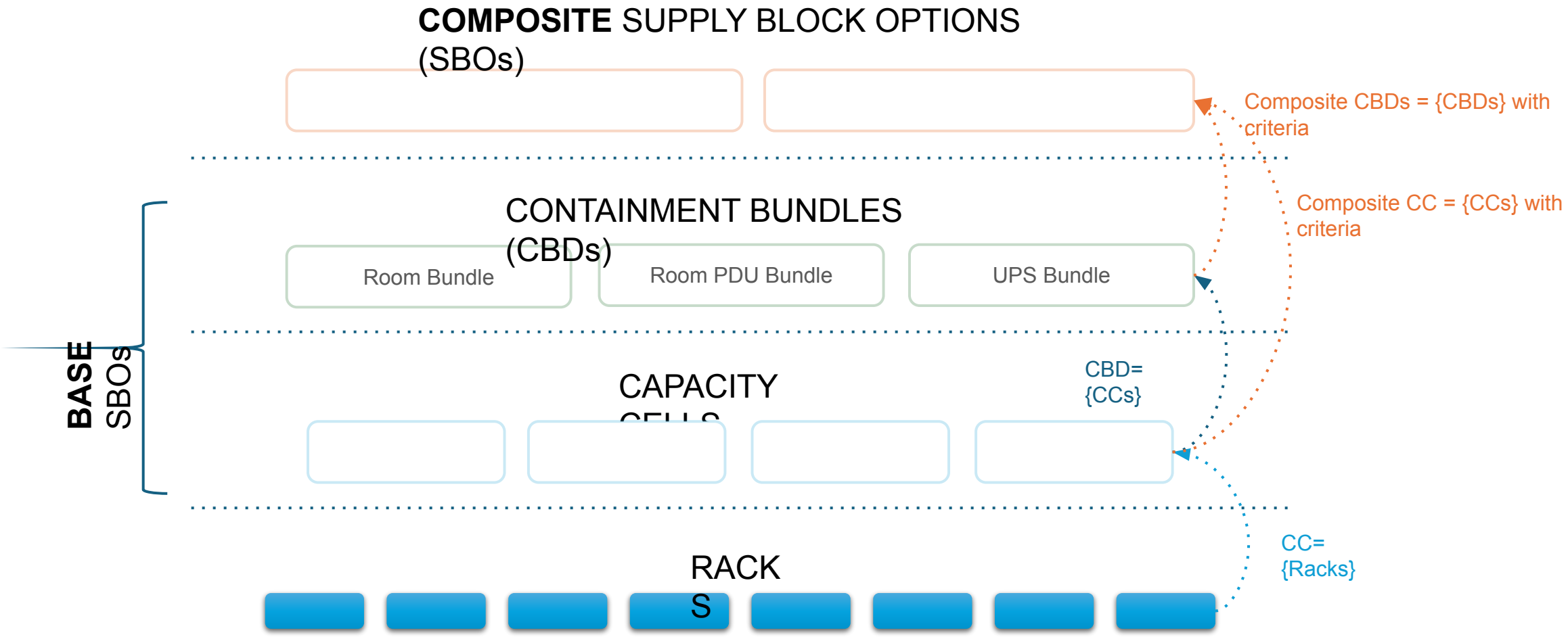
Composite Supply Block Options: certain CCs and CBDs can be combined together **within their own types** to form composites, IF they meet the composition rules. These composites are called **Composite SBOs** *(Composite CC, Composite Room Bundle, Composite Room PDU Bundle, Composite UPS Bundle).*

Only Base SBOs are reflected in the whitespace topology. Composite SBOs are separately recorded.

The Composition Rules and the SBO_Builder requirements are here:
https://docs.google.com/document/d/1pz1vWC1ny1u_A4zKELW_zlqIdpuoVgNHedTIGBMi35U/edit?tab=t.0

Supply Block Options = Base SBOs & Composite SBOs

Base SBOs = CCs & CBDs



Whitespace Packing Process

RAPs Builder

Resolved **A**ssociation **P**aths for Racks

CC Packer

Capacity Cell Packing

CBD Packer

Containment Bundle Packing

SBO Builder

Supply Block Options Building

- Create/ update whitespace topology,
- Update composite SBO data

First, we need to have a table to easily look up how a rack is associated with rooms, RPPs, AZ, LL, Room PDU, UPS, and other related infrastructures.

This is done by the raps_builder.

Then we pack the racks into Capacity Cells.

Capacity Cell: the smallest group of racks that do not share their {AZ, LL, RPP} with any outside rack.

This is done by the cc_packer.

Then we pack CC into Containment Bundles.

Room bundle: the smallest group of CCs that do not share any room with any outside CC.

Room PDU bundle: the smallest group of CCs that do not share any room PDU with any outside CC

UPS bundle: the smallest group of CCs that do not share any UPS with any outside CC

This is done by the cbd_packer

Then we build Composite SBOs within each **Base SBO families** *(Composite CCs, Composite Room Bundles, Composite Room PDU Bundles, Composite UPS Bundles).*

Afterwards we do a **total sbo count** for each SBO family.

This is done by the sbo_builder

SBO Family: *the Base SBO and Composite SBO of the same type: CC family, Room Bundle family, RPDU Bundle family, UPS Bundle family.*

Business and Technical Requirements

RAPs Builder:
https://docs.google.com/document/d/1amzwPZmbEzjtYYaYO6CmwfudGhJkQ1WsYvZVe5-_CkY/edit?tab=t.0

- *The RAPs builder can be used to find association path for any node type. In this case we use it for the Rack.*
- *ss*

CC Packer: https://docs.google.com/document/d/1H_jlgfG1P1kgJusW4TrdaaBnfSKQbF6pNoDlRZ8_wUA/edit?tab=t.0

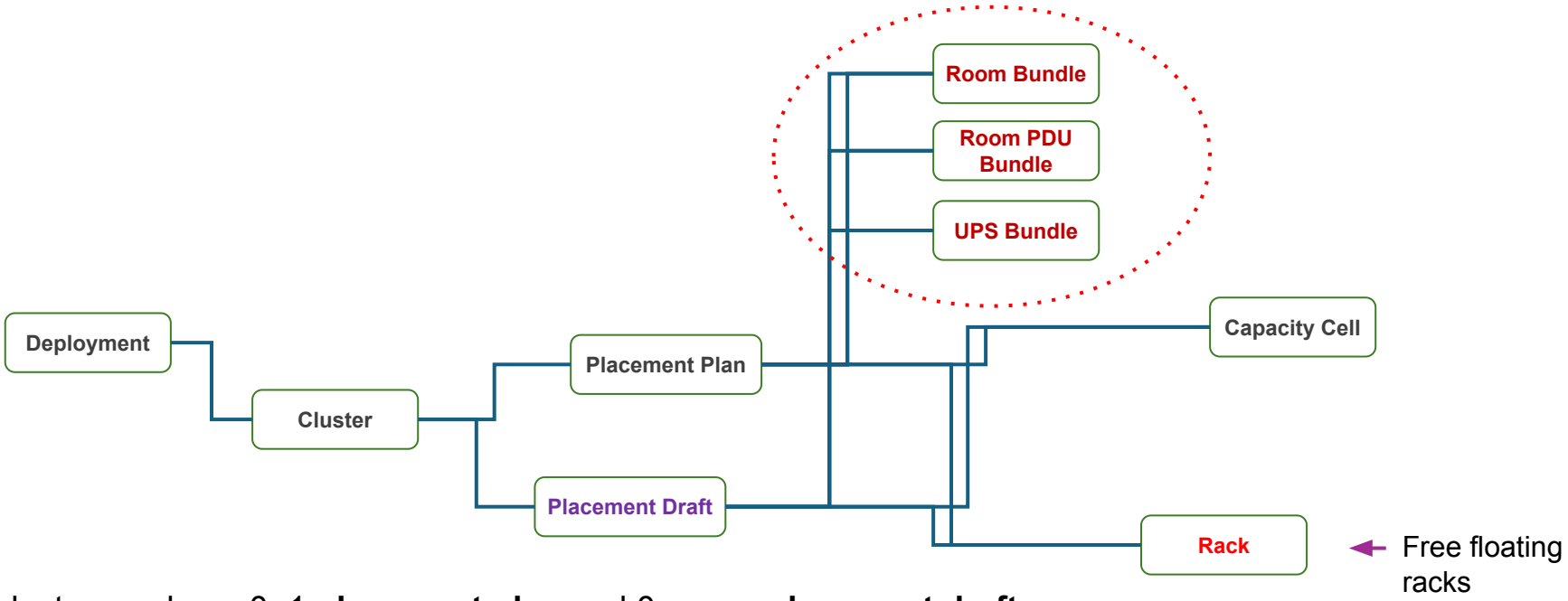
CBD Packer: https://docs.google.com/document/d/1Aiu91HzFb3uiVfHwrQslLfMvb_HhFQ9DNDPayAy_65M/edit?tab=t.0

SBO Builder:
https://docs.google.com/document/d/1pz1vWC1ny1u_A4zKELW_zlqIdpuoVgNHedTIGBMi35U/edit?tab=t.0

Deployment Planning

Deployment Blueprint Update

A placement can now directly use Containment Bundles, in addition to CCs and Racks



- A cluster can have 0~1 **placement plan** and 0~many **placement drafts**
- A placement plan/ draft include **Capacity cells** and (sometimes) **"free-floating" racks, room bundles, room PDU bundles, UPS bundles.**

What is **"Free-Floating" Racks?**

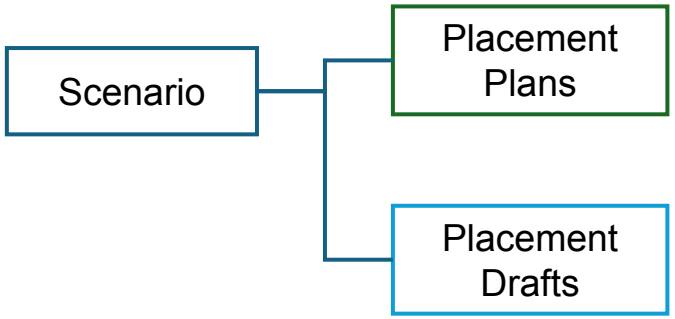
In the Whitespace Blueprint, all racks belong to and belong to only one Capacity Cell.

When building Placement Plans, when the Capacity Cell is used, all the racks of the Cell will be associated with the Plan.

However, occasionally the planner may choose only to use individual racks of a Cell (this breaks the integrity of the Cell). So when racks are directly associated the the Placement Plan, they are called "free-floating" racks.

Scenario Model

We introduce the concept of **Scenarios** to manage data that need to have different values depending on the specific Placement Draft. **Scenario is needed in most Deployment Blueprint related data and should be established as a core construct.**



General Definition: A scenario is a “reality” where certain subset of placement plans and/or placement drafts are in effect/ taken-into-account.

The following data are all Scenario-Dependent:

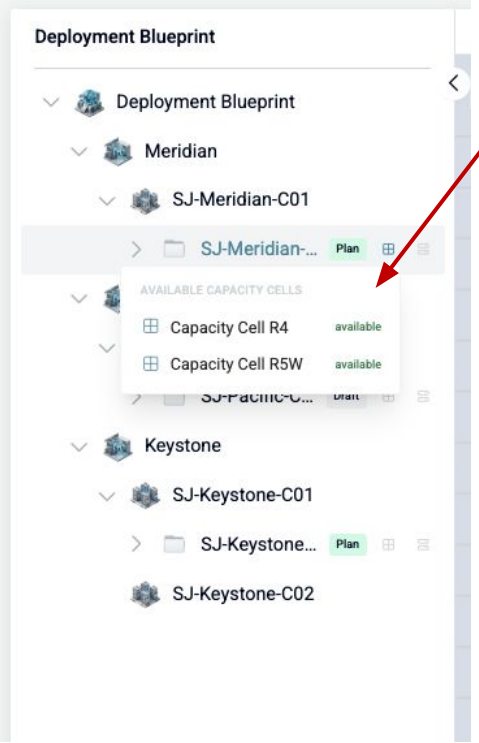
- All load and margins in the Load Capacity Calculator
  - Task: the different sets of values in the Load Capacity Calculator should be aligned with the Scenario Model
- Deployment reservations
- Most variables in the Optionality Value Calculator (aka. OAKw Evaluator)

Scenarios (scenario instances)

To start with (version 1), we adopt a simplified set of Scenarios

1. One Single **Planned Scenario (S0)**: the scenario where all the placement plans are included (accounted for) and NO placement draft is included. (one cluster can only have one PP) – this is the "hardened" reality, not "what-ifs".
2. One Single **Whitespace Scenario (S_W) :** the scenario where NO placement plans or drafts is accounted for – basically everything is unused.
3. Multiple Draft Scenarios **$S_{(PD)}$**: **each placement draft has a its own draft scenario**. In a draft scenario all the placement plans of other clusters are included, **as well as** this placement draft. If this same cluster has a placement plan, it is NOT included. If this cluster has any other placement drafts, they are NOT included. The $S_{(PD)}$s represent "what-if" realities.

Deployment Reservations



In the Deployment Blueprint, we “assign” CC or Racks to Deployment Plans or Drafts before any load can be placed on their racks.

This is called “reservation”. Reservations must follow certain rules, so that the same structure doesn’t get reserved twice in conflict.

Any reservation must be associated with a specific Placement Plan or Placement Draft.

UI task: we need to expand reservable entities to include also Containment Bundles (in addition to racks and CCs). But we will NOT support directly reserving Composite SBOs at this point.

Reservation Status (RST)

”Reservable Entities” - Racks and Base SBOs (CCs and CBDs) can be directly reserved by the user.
Reserved Racks and base SBOs can also be “released”.
Reservation associates the rack and or SBOs to a specific Placement Plan or Draft.
Reservation and release can happen both to Placement Drafts, or directly to Placement Plans
Composite SBOs are not directly reserved. But the user can separately reserve all components of a Composite SBO. In the backend the reservation status of Composite SBOs are still computed.

Reserved: the rack or the entire SBO, including all its sub-elements, are reserved to a **single PP or PD.**
For example,
- *if a CC is reserved, all its racks are no longer free.*
- *If a Containment Bundle is reserved, all its CCs and racks are no longer free*

***Reserved* ≠ Rack Load Allocated**. Reservation status is an assignment by the block. Even when no rack load has been allocated, the base SBO can still be reserved.

Partially Reserved: when **some but not all** of a SBO’s racks are reserved to a particular PP or PD, the SBO is “partially reserved”. When all of a SBO’s racks are reserved, but reserved to different PPs or PDs, it is also ”partially reserved”.
For example,
- *If CC01 is reserved, the Room Bundle that contains CC01 will be partially reserved.*
- *If a rack in CC01 is reserved, the Room Bundle that contains CC01 will also be partially reserved.*
- *If CC01 is reserved to Cluster-A-PP, CC02 is reserved to Cluster-B-PP, the Room Bundle that contains CC01 and CC02 will be “partially reserved”.*

Free: no rack or sub-element in the SBO is reserved or partially reserved.

Reservation Statuses are Scenario-Dependent

Reservation Conflicts

Since we allow different scenarios (especially $S_{(PD)}$s), a same rack or SBC could be ”double-booked” between two PDs, or between a PP and another PD. (The ”double-booking” should never happen between two PPs.)

With the use of different scenarios, such double-booking (or reservation conflicts) is tolerated as long as it happens on Placement Drafts.

However, the Reservation Conflicts need to be tracked, because the gate the conditions before a Placement Draft can become Placement Plans (conflicts should not be allowed to spill into Placement Plans).

Track fundamental Conflicts at the rack level. The conflicts at SBO levels are derivatives.

Reservation Conflict Status (RCST) is scenario-specific

- Hard Conflict: the rack reserved in this PD is also reserved by another PP
- Soft Conflict: the rack reserved in this PD is also reserved by one or more other PDs
- None: (1) no conflict for a rack in a PD; or (2) the rack reserved in this PP is also reserved by other PDs, but since PP’s dominate over PDs, the conflict can be ignored by the PP.

PDs with Hard Conflicts cannot be turned into PPs before the conflicts are resolved or turned into soft conflicts

PDs with Soft Conflicts can be turned into PPs. Doing so will change some other PDs conflict from Soft Conflict into Hard Conflicts

(1) Updating RST and RSCT upon Reservation

1. Validation before reservation

Only racks and base SBOs whose RST = *free* in the S_0 (planned scenario) can be reserved.

Racks and base SBOs that are free in S_0 but not free in $S_{(PD)}$ can still be reserved.

2. Direct status change

Associate the reservable entity with the Placement Plan or Draft in the deployment blueprint topology.

Change the reservable entity's RST ***reserved*** in the appropriate scenario (S_0 or one of the $S_{(PD)}$'s)

3. Update rack RST

For all the racks within the reserved entity in #2, update their Rack.RST ***reserved*** in the appropriate scenario (S_0 or one of the $S_{(PD)}$'s)

No need to update the Rack.RST in other $S_{(PD)}$ scenarios.

4. Reconcile RST for all associated SBOs.

Find ALL the SBOs (CC, CBD, composite CC, composite CBD) that are associated with any racks updated in #3:

If reservation was for a PP, **update their RST in S0:** Check if all Racks within the SBO are now reserved to the same PP. If so, update SBO.RST = reserved.
If not, update SBO.RST = partially reserved.

If reservation is for a PD, **update RST in $S_{(PD)}$**, same as above

5. Update Reservation Conflicts

For all the racks within the reserved entity in #2, check if they have been reserved more than once across all PPs and PDs.
(this should not happen across PPs, but is entirely possible between PP and PDs, and between PD and PDs)

a) If a Rack in a PD is reserved > once and it conflicts with a PP, mark the rack's RST_Conflict = "hard conflict" in this specific $S_{(PD)}$

b) If a Rack in a PD is reserved > once and it conflicts with other PDs, mark the rack's RST_Conflict = "soft conflict" in this specific $S_{(PD)}$

c) If a Rack in a PP is reserved > once, it must be conflicting with PDs (not PPs). No need to update the RST_Conflict value in S_0. It should remain "none".

Updating RST and RCST after Release from Placement Plans

1. Validation before release

Only racks and base SBOs associated with the PP can be “released” from the PP.

Their RST should be “reserved” , or “partially reserved” before release.

2. Direct status change

Disassociate the released entity from the Placement Plan in the deployment blueprint topology.

Change the released entity’s RST ***free*** in S0

3. Update rack RST

For all the racks within the released entity in #2, update their Rack.RST ***free*** in S0

4. Reconcile RST for all associated entities.

Find all the SBOs (CC, CBD, composite CC, composite CBD) that are associated with any racks updated in #3:

Update their RST in S0:
Check if any racks within the SBO are still reserved to ANY PP. If so, update SBO.RST = partially reserved.
If no racks is reserved to any PP anymore, update SBO.RST = free.

5. Update Reservation Conflicts

For all the racks within the released entity in #2, check if their RST_Conflict should be updated in any $S_{(PD)}$ due to the release. If so update in the respective $S_{(PD)}$

Updating RST and RCST after Release from Placement Drafts

1. Validation before release

Only racks and base SBOs associated with the PD can be “released” from the PD.

Their RST should be “reserved” , or “partially reserved” before release.

2. Direct status change

Disassociate the released entity from the Placement Draft in the deployment blueprint topology.

Change the released entity’s RST ***free*** in $S_{(PD)}$

3. Update rack RST

For all the racks within the released entity in #2, update their Rack.RST ***free*** in $S_{(PD)}$

4. Reconcile RST for all associated entities.

Find all the SBOs (CC, CBD, composite CC, composite CBD) that are associated with any racks updated in #3:

Update their RST in $S_{(PD)}$:
Check if any racks within the SBO are still reserved to ANY PP or THIS PD. If so, update SBO.RST = partially reserved. If no racks is reserved to ANY PP or THIS PD anymore, update SBO.RST = free.

DO NOT need to check if any rack is still reserved to any OTHER PD – because it will not affect the RST in $S_{(PD)}$

5. Update Reservation Conflicts

For all the racks within the released entity in #2, check if their RST_Conflict should be updated in any $S_{(PD)}$ due to the release. If so update in the respective $S_{(PD)}$

Turning a Placement Draft into Placement Plan

Validation: no hard reservation conflicts

Updates needed:

- All related S_0 parameters need to be re-computed, including load, margins, RSTs, RCSCs, and optionality (OAKw) values.

- All related other $S_{(PD)}$ parameters also need to be re-computed, including load, margins, RSTs, RCSCs, and optionality (OAKw) values.

Optionality Value Calculator

(aka. OAKw Evaluator)

Infrastructure Containment

Data Centers are like real-estate in that they are committed to tenants in **blocks.**

Containment:
- Tenants want to have “carved-out” infrastructure block that is “not shared” with other tenants.
- Even enterprises that use the entire data center would prefer containment of their special projects (isolate one type of project against other projects)
- Containment creates operational predictability, reliability, accountability, and ease of management

Containment Boundaries:
- in the past, the most common containment boundary is the rack.
- This is no longer sufficient in the age of AI, because the thermal and power stress are becoming more prominent.

Capacity Cell (CC):
- In our model, the Capacity Cell is the most basic containment boundary beyond the rack.
- A Capacity Cell is a set of racks who do not share any Air Zone, Liquid Loop, or RPP/Busway Segment with any outside racks.
- The Capacity Cell achieves a reasonable level of power and thermal isolation.
- In addition, due to the Air Zone isolation, racks within the CC are usually spatially contiguous, making them more adaptable to the network configuration as well.

Containment Bundles:
- Multiple Capacity Cell that achieve additional levels of isolation together.
- Room Bundle {Room, AZ, LL, RPP/BWS}; Room PDU Bundle {Room PDU, AZ, L, RPP/BWS}, UPS Bundle {UPS, AZ, LL, RPP/BWS}

Optionality Adjusted Kw Value (OAKw)

A physically available kW is more valuable when it can participate in more commercially meaningful “future shapes”. This is called **“Optionality”.**

These future shapes are the Capacity Cells and Containment Bundles.

The **OAKw Evaluator** calculates the OAKw value of each kW, and for the whole facility.

OAKw value is also scenario-dependent.

OAKw can be pronounced as
“Oak”

1. CC-Allocated Capacity (CCAC), for Row and Rack

Although a Rack has its Rack_Circuit_Capacity_(design), the actual usable Rack kW may still be limited by the RPP, AZ, or LL that serves it.
A Row usually does NOT have its intrinsic (native) kW capacity value.

***CCAC** is a portion of the **CC Capacity Envelope** that is logically allocated to any unit below the CC level for computational purposes. It does not usually carry a real physical constraint or allocation.*

CC-Allocated Row Capacity (Row.CCAC) is a portion of the CC Capacity Envelope allocated to the row. CC-Allocated Rack Capacity (Rack.CCAC) is a portion of the CC Capacity Envelope allocated to the Rack.
(In physical reality, it is possible for a Row or a Rack's load to exceed its CCAC or be capped lower at its CCAC)

$$\text{Row.CCAC} = CC.Capacicy\ Envelope \times \frac{Row.Total_Rack_Circuit_Capacity_(design)}{CC.Total_Rack_Circuit_Capacity_(design)}$$

$$\text{Rack.CCAC} = CC.Capacicy\ Envelope \times \frac{Rack_Circuit_Capacity_(design)}{CC.Total_Rack_Circuit_Capacity_(design)}$$

2. Row Quality Profile, Row Quality Premium

Quality means the design characteristics of the data center infrastructure that makes it more or less commercially attractive. Strong qualities add commercial premium (more expensive).

Quality can be characterized at different levels: rack, row, room, etc. Currently we select Row as the most basic quality unit. **Quality Premium** is a composite score based on 4 predefined Profile Attributes, and possibly also incorporate Custom Factors.

Row Profile Attributes	Tiers	Premium value/ Weight
1. Density Bracket *(per rack)*		**35%**
*(*use average if each rack is different)*	<= 15 kW/ rack	1.0
	15 – 30 kW	1.05
	30 – 60 kW	1.12
	60 – 120 kW	1.22
	>= 120 kW	1.30
2. Power Redundancy Tier		**20%**
	N (single feed)	1.0
	N+1	1.03
	A/B feeds	1.08
	2N architecture	1.14

Row Profile Attributes	Tiers	Premium value/ Weight
3. Cooling Redundancy Tier		**15%**
	N	1.0
	N+1	1.03
	N+2	1.08
	N+N (dual loop)	1.14
4. Cooling Mode		**30%**
	Air Only	1.0
	Air + RDHx	1.03
	Air + DTC	1.07
	Full Liquid DTC	1.07
Custom Factor		

$$\text{Row Quality Premium} = \sum_{all\ Attributes} Attribute\ Value \times Attribute\ Weight$$

3. Quality-Weighted Value for Row (Row.QV)

Reference kW Rate: the baseline $ value of an average 1 kW at the data center
(this should be user configured, for **default use $180/kW**)

Row.QV = Row.CCAC × Reference kW Rate × Row Quality Premium

4. Capacity Cell Coherence Factor (CF)

The Capacity Cell Coherence Factor measures **how internally uniform the rows are across the 4 quality dimensions.**

Within each Row profile attribute:

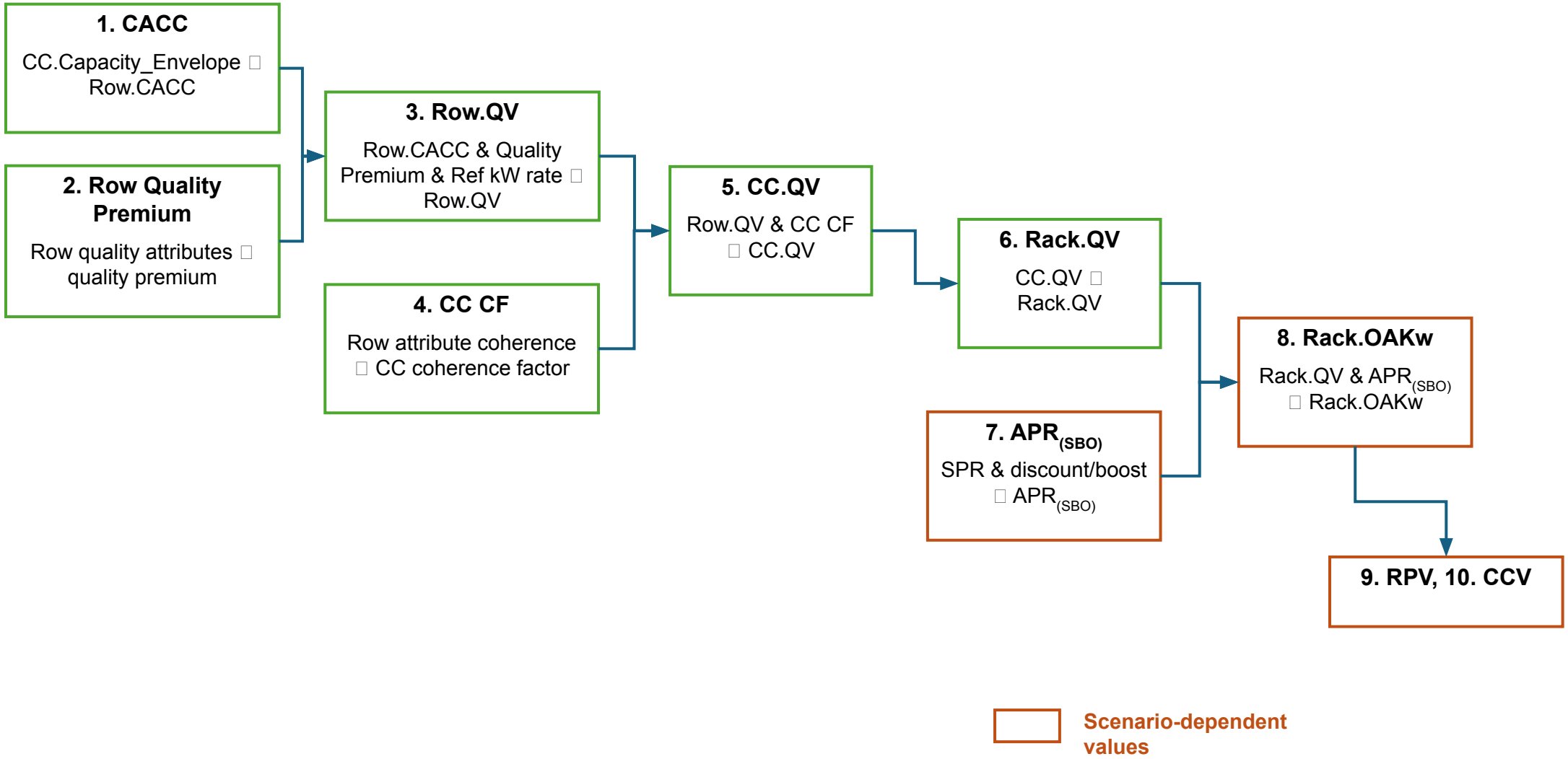
Distribution of attribute value	Attribute Coherence Factor
All rows are of the same tier	1
Two different tiers exist across rows	0.98
Three or more different tiers exist across rows	0.92

Weight of each attribute

Row Profile Attributes	Weight
Density Bracket	35%
Power Redundancy Tier	20%
Cooling Redundancy Tier	15%
Cooling Mode	30%

$$\text{CC Coherence Factor} = \sum(Attribute\ CF \times Attribute\ Wt)$$

Summary of OAKw Process



5. Quality-Weighted Value – Capacity Cell (CC.QV)
6. Quality-Weighted Value – Rack (Rack.QV)

CC.QV = $\sum Row.QV$ **× Capacity Cell Coherence Factor**

Add up all the Row.QV of the CC, then multiply by the Coherence Factor.

Rack.QV = $CC.QV \times \frac{Rack_Circuit_Capacity_(design)}{CC.Total_Rack_Circuit_Capacity_(design)}$

Allocate the CC.QV to the Rack by proportion of the Rack_Circuit_Capacity_(design)

7. SBO Membership Premium

If a Rack belongs to a Base SBO, it will receive a premium on top of its Rack.QV.

Each Base SBO has a **Standard Premium Rate (SPR)** (of *x%) and an* ***Adjusted Premium Rate (APR)***

- Composite Boost: If a Base SBO participates in additional Composite SBOs, then the Base SBO's **premium rate will be boosted** (increased).
- Entanglement Discount: If a base SBO happens to cross certain containment boundaries (for example, spanning over multiple rooms), then this base SBO's **premium rate will be discounted (decreased).**

SPR x any applicable boost or discount x *(RST effect)* = APR

Reservation Status (RST) effect: base SBOs that are partially reserved loses its premium.

APRs are scenario-dependent: because APRs has an RST effect, and RST are scenario-dependent, therefore expect the APR in each scenario to be different.

More details on APR, boost, discount, the the RST effect will be covered later

From Standard Premium Rate (SPR) to Adjusted Premium Rate (APR)

The APR Calculator calculates APR for each base SBO, using SPR and Base SBO associations

APR is scenario-dependent: reservation status affects APR

Base	Standard Premium Rate (SPR)	Entanglement Discount	Composite Boost	RST Effect
Capacity Cell	5%			If the CC is partially reserved, APR = 0
Room Bundle	2%			If the Room Bundle is partially reserved, APR = 0
Room PDU Bundle	3%			If the RPDU Bundle is partially reserved, APR = 0
UPS Bundle	3%			If the UPS Bundle is partially reserved, APR = 0

Inputs Used to Calculate APR for each specific SBO and specific Scenario

The SBO_Builder generates a list of Composite SBOs.
It also outputs a *sbo_association_symmary* like below. The total_sbo_count here is the right count for the Whitespace Scenario (S_W)

base_sbo_id	base_sbo_type	self_sbo_count	composite_sbo_count	total_sbo_count
CCS-000004	CC	1	1	2
CCS-000005	CC	1	1	2
UPB-000001	UPS_BUNDLE	1	1	2
UPB-000002	UPS_BUNDLE	1	1	2
CCS-000001	CC	1	0	1
CCS-000002	CC	1	0	1
CCS-000003	CC	1	0	1
RPB-000001	ROOM_PDU_BUNDLE	1	0	1
RPB-000002	ROOM_PDU_BUNDLE	1	0	1
RPB-000003	ROOM_PDU_BUNDLE	1	0	1

In the Deployment Blueprint (S_0 and $S_{(PD)}$), racks and SBOs can be reserved. As a result, any SBO can show "partially reserved" status.

- If a base SBO is "partially reserved", then APR = 0 (even the SPR no longer apply)
- If a base SBO is either "free" or "reserved", the SPR still applies. However, its *total_sbo_count* may be changed due to reservations in the specific scenario. A "partially reserved" composition SBO can no longer participate in the total_sbo_count. Therefore a scenario-specific total_sbo_count needs to be computed, and used to calculate APR.

8. Rack OAKw

Rack.OAKw = Rack.QV × (1+ APR_{cc}) × (1+ $APR_{RoomBundle}$) × (1+ $APR_{RPDUBundle}$) × (1+ $APR_{UPSBundle}$)

For each rack, identify its CC, Room Bundle, RPDU Bundle and UPS Bundle, use their APRs in the above formula

Notes

(1) as per definition of CC and Containment Bundles, each Rack can only belong to one Base SBO of the same type

(2) The APRs only exist for the Base SBOs. The additional "benefits" of belong to Composite SBOs are already accounted for via the SPR to APR process

Rack.OAKw are scenario-dependent, because APRs are scenario-dependent.

9. Remaining Portfolio Value (RPV)

RPV is the total option-adjusted kW value of the parts of IT_Facility that has not yet be planned or committed.

RPV is calculated bottom-up using Rack.OAKw.

Because Rack.OAKw is scenario-dependent, RPV is also scenario-dependent.
For the **Whitespace Scenario**

- Remaining Portfolio Value = IT_Facility.OAKw = total of Rack.OAKw in the facility.
- Use S_w APRs to calculate Rack.OAKw

IT Facitly.OAKw Rate = IT Facility.OAKw/ Facility_Deployable_ITCapacity

For the **Planned Scenario S_0**

- Remaining Portfolio Value = total of Rack.OAKw from **free** racks in S_0
- Free racks: racks whose RST = free.
- Use S_0 APRs to calculate Rack.OAKw

For the **Draft Scenarios $S_{(PD)}$**

- Remaining Portfolio Value = total of Rack.OAKw from **free** racks in $S_{(PD)}$
- Free racks: racks whose RST = free.
- Use $S_{(PD)}$ APRs to calculate Rack.OAKw

10. Captured Capacity Value (CCV)

CCV is the total option-adjusted kW value of the racks reserved for the cluster or a whole deployment (multiple clusters)

CCV is calculated bottom-up using Rack.OAKw.

Because Rack.OAKw is scenario-dependent, CCV is also scenario-dependent.
For the **Planned Scenario S_0** (Placement Plans)

- CCV = total of Rack.OAKw from **each reserved racks in each placement plan**
- Reserved racks: racks reserved to this Placement Plan.
- Use S_0 APRs to calculate Rack.OAKw

For the **Draft Scenarios $S_{(PD)}$**

- CCV = total of Rack.OAKw from **each reserved racks in each placement draft**
- Reserved racks: racks reserved to this Placement Draft.
- Use $S_{(PD)}$ APRs to calculate Rack.OAKw

SJ-Meridian-C01 / SJ-Meridian-C01-PP01

kW Placed / Requested

450kW / 450kW 100%

Racks Placed / Requested

36 / 36 100%

Captured Capacity Value	Average CCV/ Rack	Remaining Portfolio Value (optionality adjusted)	Optionality Impact
$778,140	$43,230	$24,918,438	$ (31,126)

Total kW Requested	Total Racks Requested	Cooling Mode	Power Redundancy
450 kW	36	Air, RDHx	N

Rack Role Placement

Role	Target kW	Count	Placed	Cooling	Placement
Compute	15 kW	18	18	↔ Air, RDHx	CC-R1R2
Compute	10 kW	18	18	↔ Air, RDHx	CC-R1R2

Capacity Cells

CC-R1R2

920 kW guaranteed · 36/48 racks · 450 kW allocated

Power Load/ Power Capacity	Thermal Load/ Cooling Capacity
450/920 kW	450/940 kW

Power: RPP-R1-1, RPP-R1-2, RPP-R2-1, RPP-R2-2
Thermal: AIRZONE-R1, AIRZONE-R2, LL-R1, LL-R2

View Placement | Modify Placement | Run Auto-Placement | Commit Plan

Captured Capacity Value = $\sum Rack.QV$

Average CCV/ Rack = CCV/ # of Racks

Remaining Portfolio Value = R_OAKw

Optionality Impact = $\sum Rack.R_OAKw_{before} - \sum Rack.R_OAKw_{after}$

Where “before” is the scenario where the racks in this Placement Plan has not been placed.
“After” is the scenario where the racks in this Placement Plan has been placed.

System	Redundancy Option	Meaning	Typical Behavior
Power	**N (single feed)**	One electrical path to the rack.	No redundancy; failure of the path or upstream components causes outage.
Power	**N+1**	Single electrical path, but upstream equipment (UPS modules, transformers, generators) has one spare unit.	Survives one component failure but not a full path failure.
Power	**A/B feeds**	Rack has two independent power inputs from separate PDUs/RPPs and electrical paths.	Each feed often carries ~50% of load; loss of one feed keeps the rack running if equipment has dual PSUs.
Power	**2N architecture**	Two completely independent electrical systems, each capable of supporting 100% of the IT load.	Often implemented with A/B rack feeds; if one entire system fails, the other can carry the full load.
Power	**2N+1**	Two full-capacity electrical systems plus an additional spare module/unit in the plant.	Extremely high availability design; rare due to cost.
Cooling	**N**	Cooling capacity exactly equals the expected heat load.	Any failure immediately reduces cooling capacity.
Cooling	**N+1**	One additional cooling unit (CRAH, CDU, pump, etc.) beyond what is required.	Most common design; one unit can fail without overheating.
Cooling	**N+2**	Two additional cooling units beyond what is required.	Higher resilience than N+1; used where thermal continuity is especially critical.
Cooling	**N+N (dual loop)**	Two cooling loops share the heat load, but each loop may not be sized for full load alone.	If one loop fails the system runs with reduced margin.
Cooling	**2N cooling**	Two independent cooling systems, each capable of removing the full heat load.	Rare due to cost and operational inefficiency.

Reclarification of Variables, Topologies, and data consolidation

Where Can Variables be Semantically Anchored?

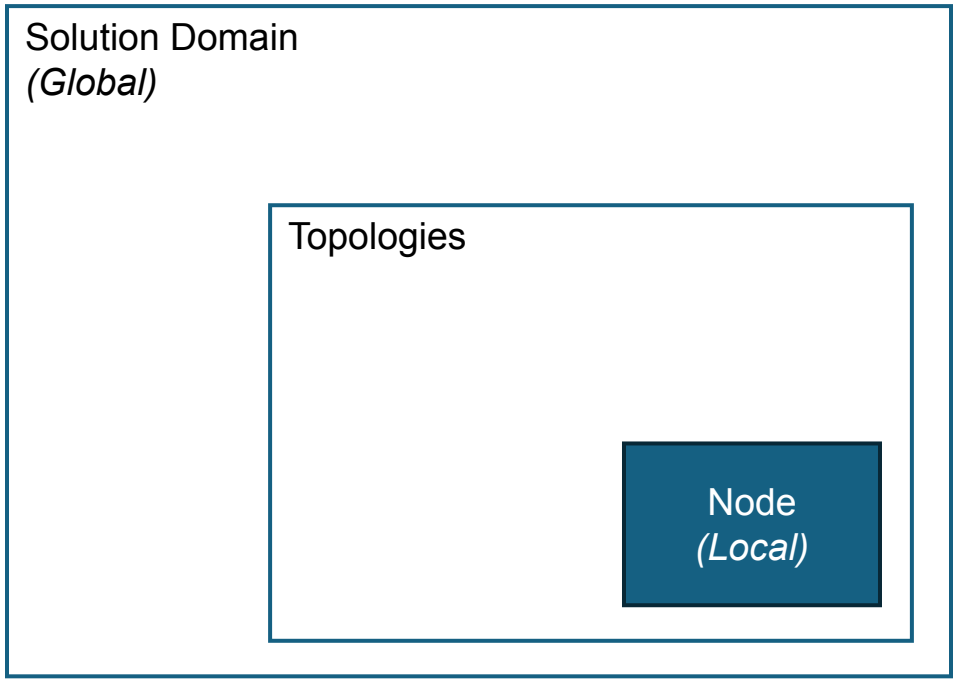
(= Where does the variable receive its **intrinsic meaning?)**

1. Globally Anchored

The variable has a global meaning, and when it is attached to any topology or node, the meaning is the same. These tend to be broad-stroke, generic concepts

Examples:

- *CO2e emission*
- *Emission activities*
- *Cost ($)*
- *Energy use*



The variable's meaning is tightly associated with an anchoring node type.

Even when the variable value is rolled-up or consolidated at a "higher" level entity, the meaning still is still attached to the anchoring node.

Example:
*Rack Circuit Capacity (RCC): if you roll up RCC to the zone or room level, they do not become the **intrinsic circuit capacity** of zone or room. They are still the just total RCC added up.*

2. Locally Anchored

3. Local Only

The variable's meaning is only associated with an anchoring node type. There is no or little meaning to try to consolidate this variable across other node types.

Examples: Server fan speed, ColdAisle Supply Temperature

In the Data Center domain, most variables are in category #2 and #3

Except for things like energy usage, emissions, power usage, water usage, and potentially gpu/cpu usage – because they are considered **"global" resources *in the context of our solution.***

Most other variables are either locally anchored, or local only.

This means a three things:

1. **Anchor Matters:** Whenever we use a variable, it has to be very clear which node type it is associated with
   - RPDU Design Capacity is not the same as UPS Design Capacity; Aisle temperature is not the same as server temperature.
   - The right practice is to prefix it with the node type: Rack.Inelt_Temperature or UPS_Design_Capacity, so we don't accidentally believe there is any connection between aisle temperature and server temperature

2. **Semantically Native or Derivative:** We need to differentiate if a variable associated with a node is semantically **Native** or **Derivative.**
   - A variable at its anchoring node directly is semantically **Native**
   - A variable whose intrinsic meaning is anchored at another original node is semantically **Derivative**. In most (but not all) cases, the values of such variables are calculated.
     - *For example: total rack capacity of a room. average rack inlet temperature in a zone.*

3. **Context Matters:** When we consolidate **Locally Anchored Variables** along topology structures to arrive at semantically derivative variable values, we need to be extra careful whether preservation of semantic meaning holds. Very often the consolidation may not be meaningful on many node levels. *For example, if you take the RPP capacity and roll up across the electrical topology, it has very limited meaning and it can even be misleading.*

Different Ways to Calculate the Values of Semantically Derivative Variables

Note: I continue to use "semantically derivative". It is meant to separate this concept from "computationally derived". A computationally derived variable value may not be semantically derivative. "Semantical" means "meaning". If we do not have Rack Power Usage data, we may derive it computationally, that does not make Rack Power Usage semantically derivative.

1. **Topology-Based Consolidation:** these are the data consolidations we do along specific topology structures. For example:
   - Consolidating rack.power_usage along the operational topology will get us total rack power use – this stands for total IT power;
   - Consolidating power_usage along the entire electrical topology will get us total facility power use, including both IT power, and power used by other cooling and lighting equipment.

2. **Group Statical Processing:**
   - Topologies here don't really matter. What matters is the Group-Member relationship. The group-member can be linked by a topology edge (e.g. row-rack: all racks in the same row, room-rack: all racks in the same room), or may have nothing to do with topology (e.g. all DTCs as a group, all GPU servers as a group)
   - The math of the processing is based on **statistics**, not structural relationships/flow.
   - Operations can be sum, average, max, min, stdev, mean, median, 95 percentile, etc.

3. **Custom logic:** the computation logic is specific to a particular node type and variable.

Note: a **standard rollup** consolidation along a topology that happens to be a **strict hierarchy** is the logical/mathematical equivalent of multiple Group Statistical Processing using SUM. An example is Emissions consolidation along the Org hierarchy – whether we do simple sum, or we follow the topology tree level-by-level is the same.

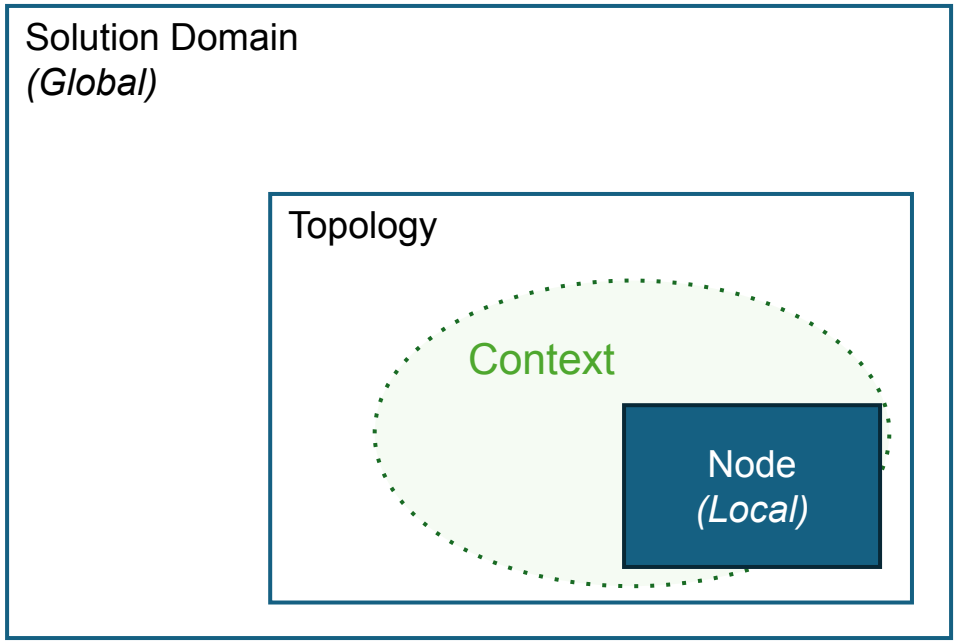
The Context Boundaries for Topology Based Consolidation

In order to closely govern topology-based data consolidation (especially for locally anchored variables), so that the computational results are aligned with the actual semantical meaning, we need to use the concept of Context.

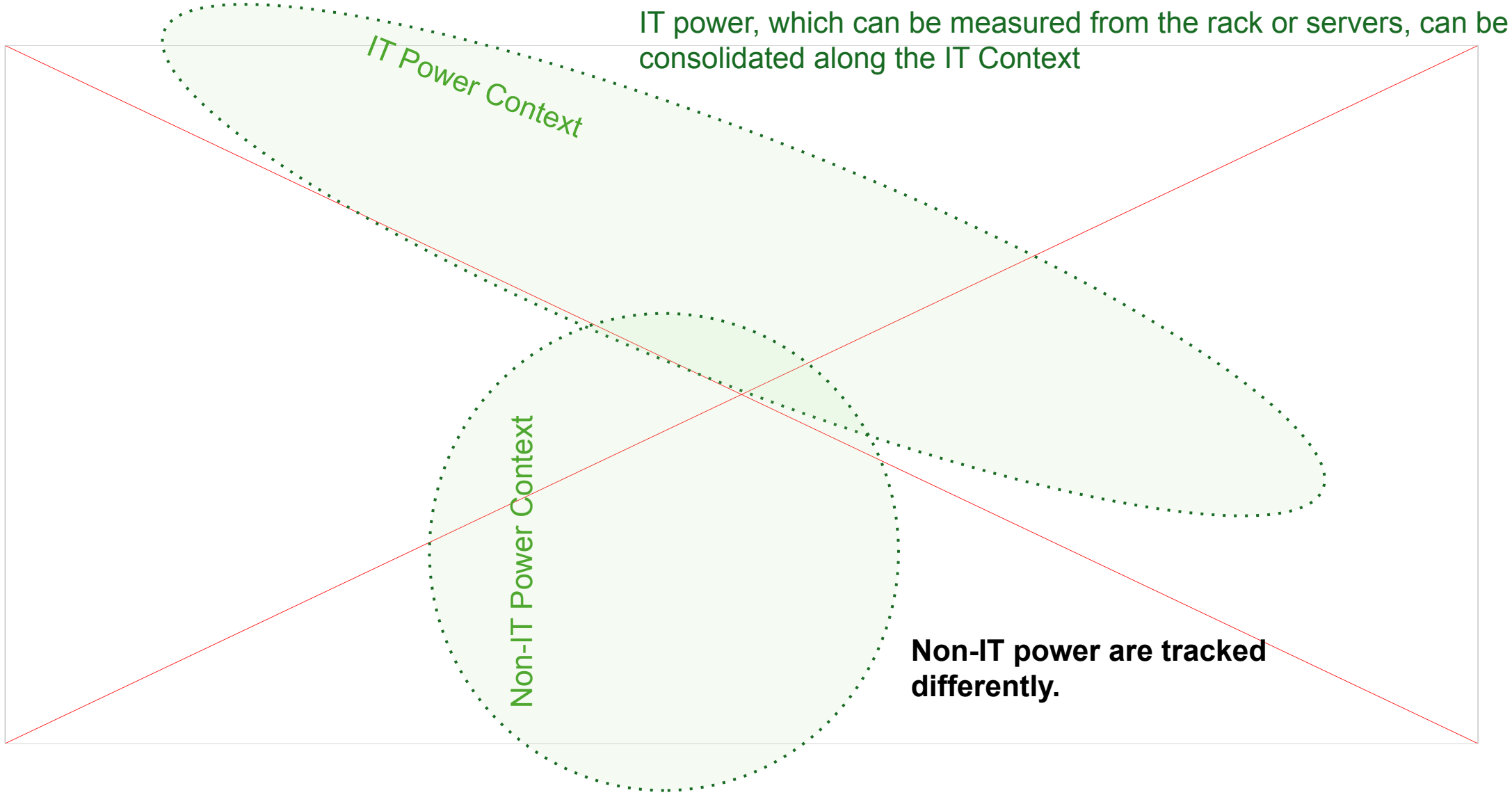
We define a **Context Scope** using the concept of **nodes and edges**. A context scope is a connected set of nodes and edges, within which certain variables can be consolidated with a consistent meaning.

Therefore:

1. A Context Scope is a sub-part of a topology, or in special cases an entire topology.
2. A Topology can have multiple Contexts. These contexts can overlap with each other.

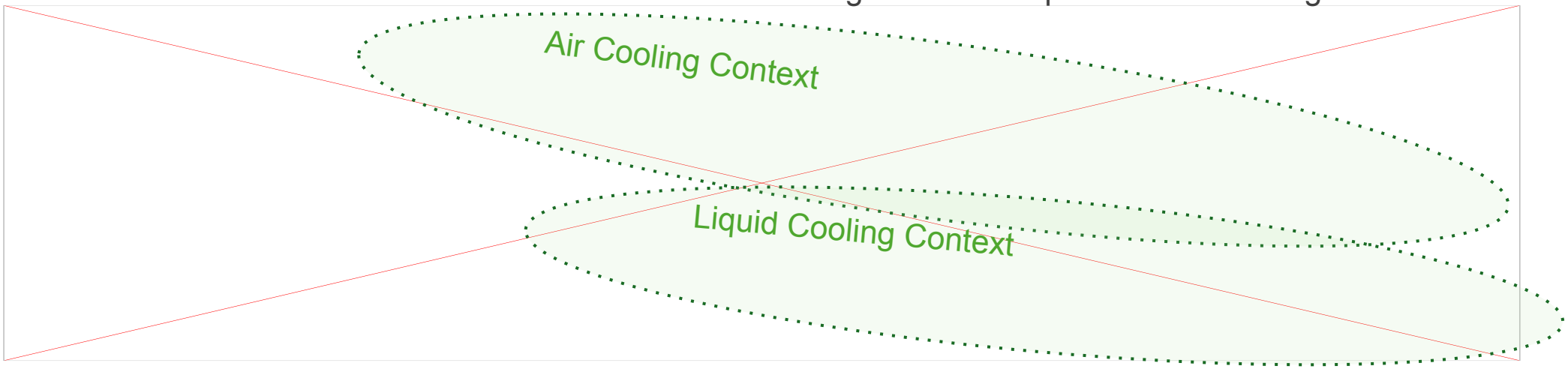


Illustrative Example of Multiple Contexts within the Electrical System



Illustrative Example of Multiple Contexts within the Cooling System

Air Cool Load can only be consolidated within the Air Cooling Context.
Trying to consolidate Air Cool Load through the liquid Cooling Context will produce misleading results



Liquid Cool Load can only be consolidated within the Liquid Cooling Context.

Summary

Variable + (Topology) Context Scope + Consolidation Logic = **Consolidation Rule**

Variables are no longer tightly tied to a single topology

Variables are defined at the Node/ Node Type level

Context will now define the scope of the consolidation, not entire topologies

Same ones:

- *Standard rollup*
- *Smart Inclusive*
- *Standard Hybrid*
- *Inclusive Flow (DAG)*

A consolidate rule will exists as an independent entity, referring to the **variable**, the **Context Scope** and the **Consolidation Logic**

How to Define Context Scope – Approach A (Simple)

Context Scope = Contextual Node/Node Type List *(Collection of Node Types & their Edge Types)*

For example, to consolidate Allocated_ITLoad from racks, these node types and the connecting edges are in scope: *Rack, Row, Zone, room, Floor, IT Facility, Site*

Pros: simple to conceptualize and define
Cons: brittle to topology changes. Any topology changes could disrupt multiple context scopes. And the context scopes need to be updated manually.

I suggest the team use Approach A first, and switch to Approach B later.

How to Define Context Scope – Approach B

Approach B involves

1) A **Declarative** Context Definition
2) A Parser that ”reads” the Context Definition and creates the Contextual Node/ Node Type list *(no longer manually created)*

Context
- Context Name

1:m

Path
- Path ID

Path Definition
- Topology: ***one or multiple*** *connected topologies*
- Inclusion Type: *Subgraph, Topology*
- Subgraph Definition: *Lineage, Branch*
- Lineage Anchor: the node type
- Branch Root: the node type
- Trim Node Type: one or more node types
- Blocked Edge Type: one or more edge types

Path: a path is a traversal path starting from a designated node type, and links all the node types together in a single traversal. There can be no disconnected nodes in path.

Path Definition Parameters:

Topology: which topologies are involved. If multiple topologies, they must be connected by portal nodes so they can be traversed in the same path.

Inclusion type: either the entire topology/topologies are included, or a subgraph only

Subgraph Definition: if subgraph, it can be defined as a Branch or Lineage. Branch means starting from a **Branch Root node** and move downstream. Lineage means starting from a **Lineage Anchor node** and move upstream.

Trim Node Type: If only a Branch Segment or Lineage Segment, then specify the trim nodes. The path will stop at the trim nodes.

Blocked Edge Type: If during path traversal downstream a certain subbranch needs to be blocked, specify the blocked edge, and all structures downstream will be excluded. If during path traversal upstream a certain parent lineage needs to be blocked, specify the blocked edge, and all structures upstream will be excluded.

Concept Clarification/ Cleanup

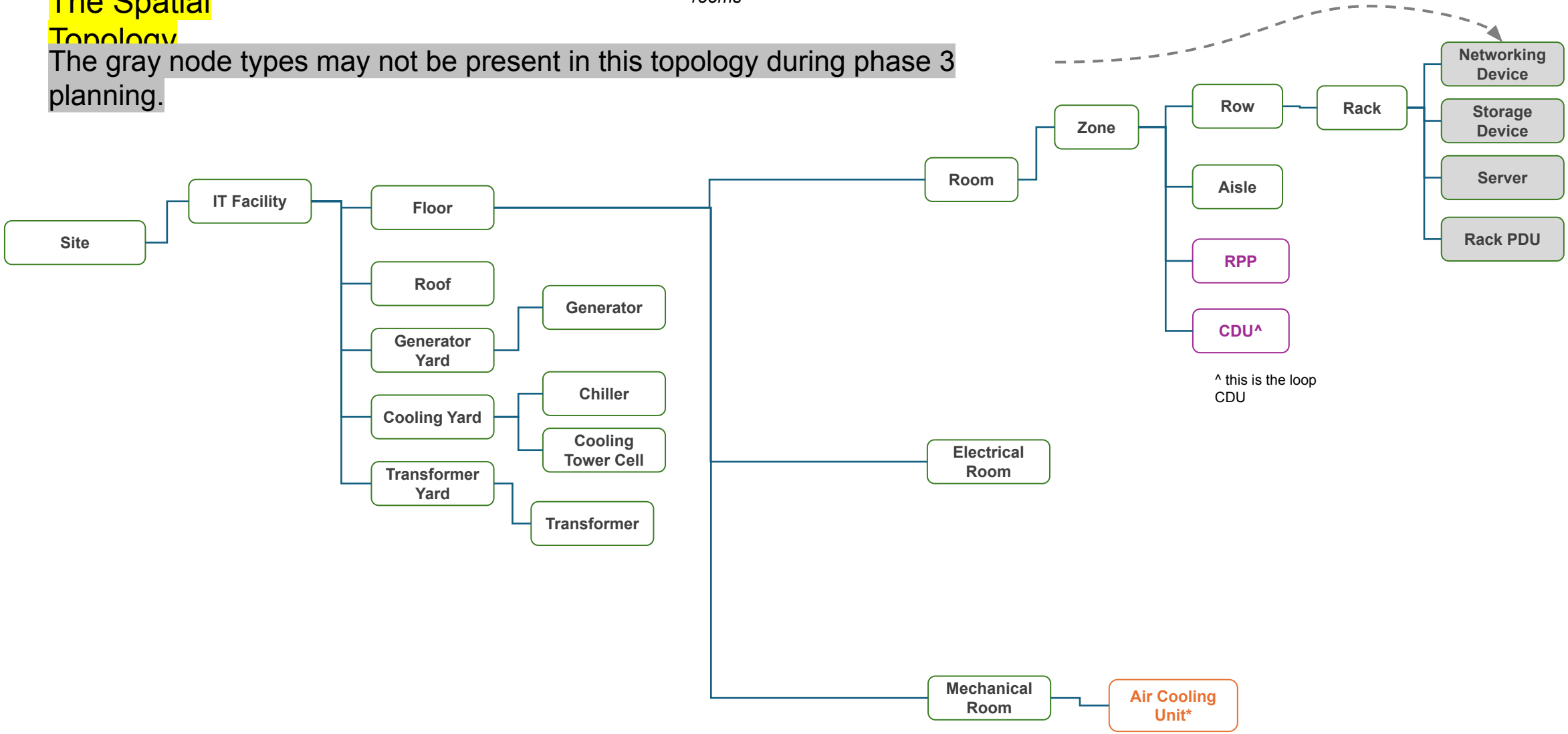
	BEFORE	NOW/ clarification
Topologies	**Primary vs. Secondary** - Only one fixed as “primary”, all others are secondary - Variables/OP are associated with a specific topology	**MANTO (navigation) vs. BENTO** - They are relative. Any topology could be MANTO, and others will be BENTO. - The distinction between primary and secondary goes away - Variables/OP are tied to nodes/ node types, not fixed at entire topology level
Topology Structures	**Hierarchy vs. DAG** - Primary topology is hierarchy. Secondary topologies can be DAG	**Hierarchy is a special case of DAG** Any topology can be DAG or hierarchy
Topology Data Consolidation	**Data consolidation logic is coupled with the topologies** - If ”primary”, has smart inclusive, standard rollup, or standard hybrid - If “secondary”/DAG, uses Inclusive Flow	**Data consolidation methods are independent of topology** - Data consolidation methods will be dependent on Context - Context + Variable specific consolidation method & logic - Context will be defined as a first-class object. A context will have its context rule to define the consolidation method
Eligibility for Data Consolidation	**Only OP in iDOT**	**Any Variable** - The existence of a consolidation method is independent from the definition of OP/Metric, and independent of topologies

Topology (WIP)

The Spatial Topology

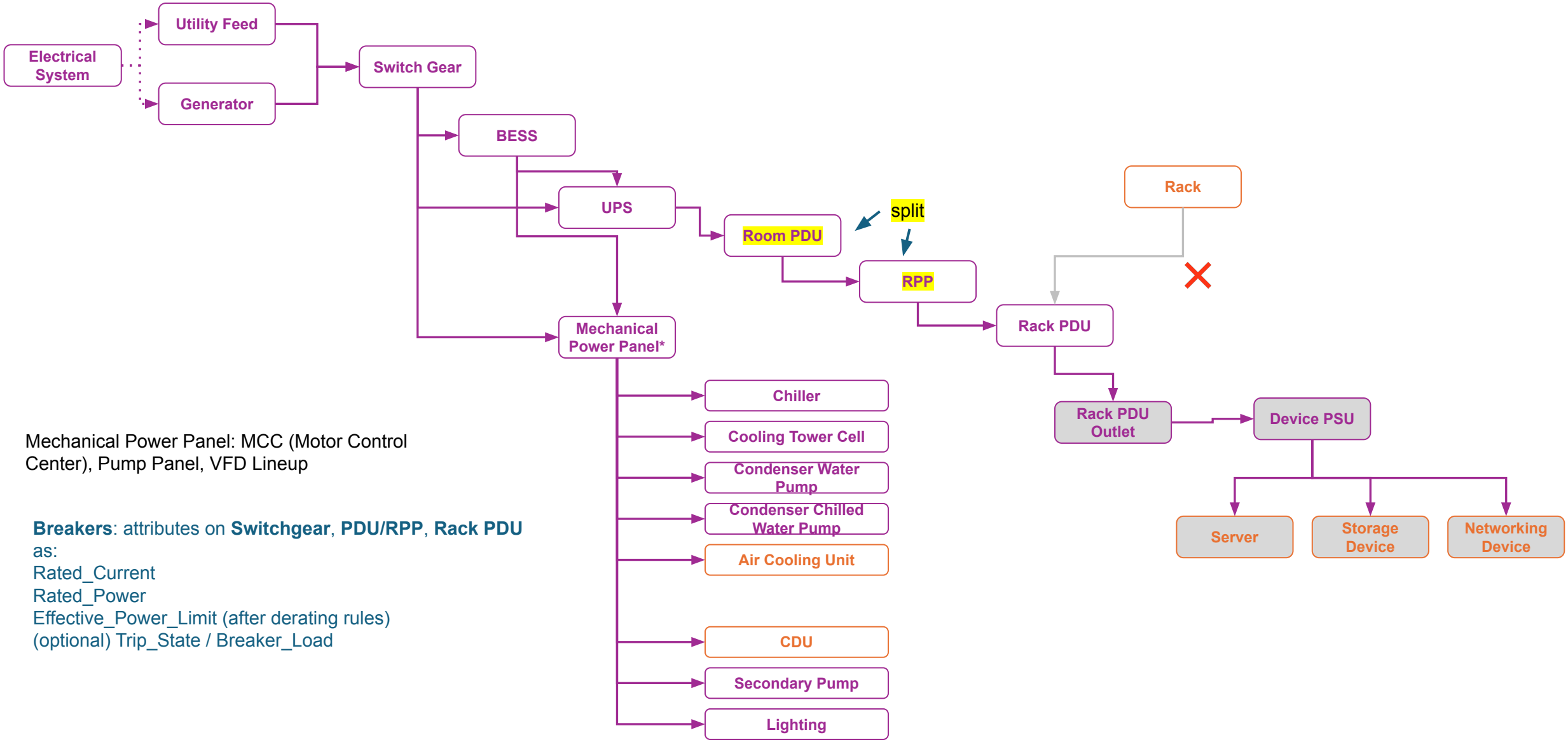
The gray node types may not be present in this topology during phase 3 planning.

** The node type "Room" will be reserved for white space rooms*



Electrical System (Multi-Context DAG)

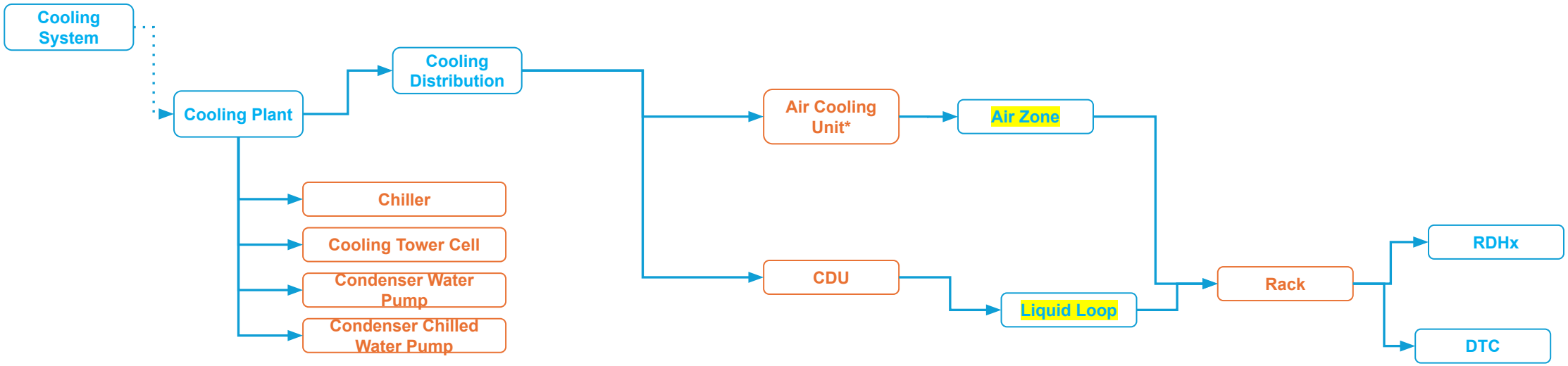
The gray node types may not be present in this topology during phase 3 planning.



Mechanical Power Panel: MCC (Motor Control Center), Pump Panel, VFD Lineup

Breakers: attributes on **Switchgear**, **PDU/RPP**, **Rack PDU** as:
Rated_Current
Rated_Power
Effective_Power_Limit (after derating rules)
(optional) Trip_State / Breaker_Load

Cooling System

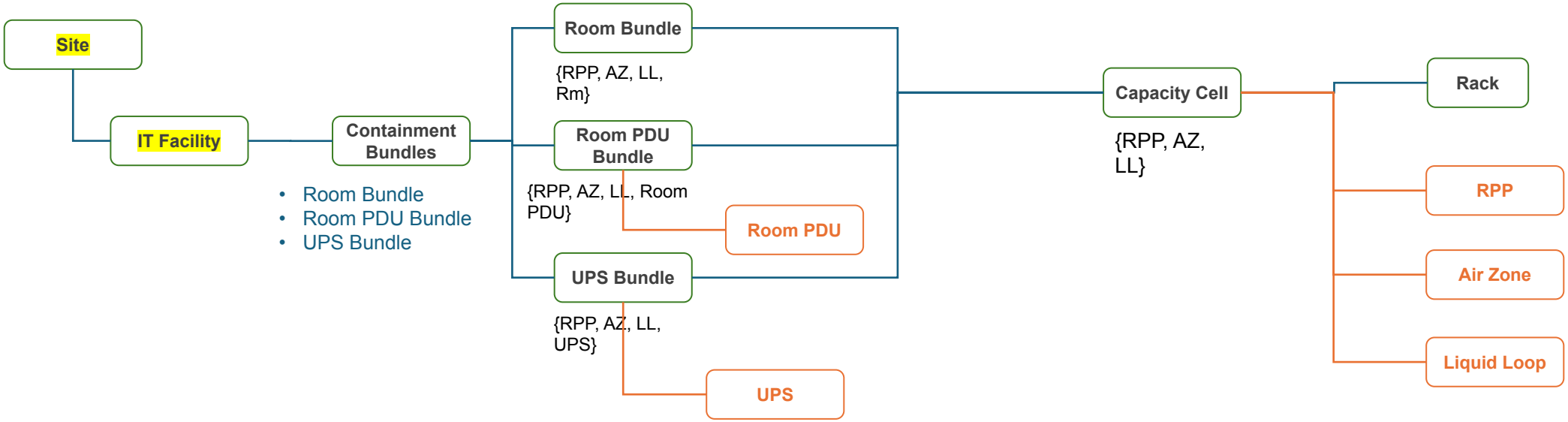


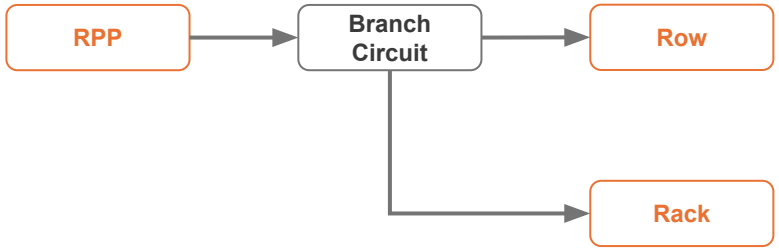
(Air Cooling Unit can be CRAH, CRAC, In-Row Air Coolers, AHU)

Central CDU is currently missing
The current CDU is the loop CDU
We will add Central CDU later, not now.

Whitespace Blueprint

The Whitespace will be ”carved” into capacity cells, then organized into containment bundles before the load placement planning starts





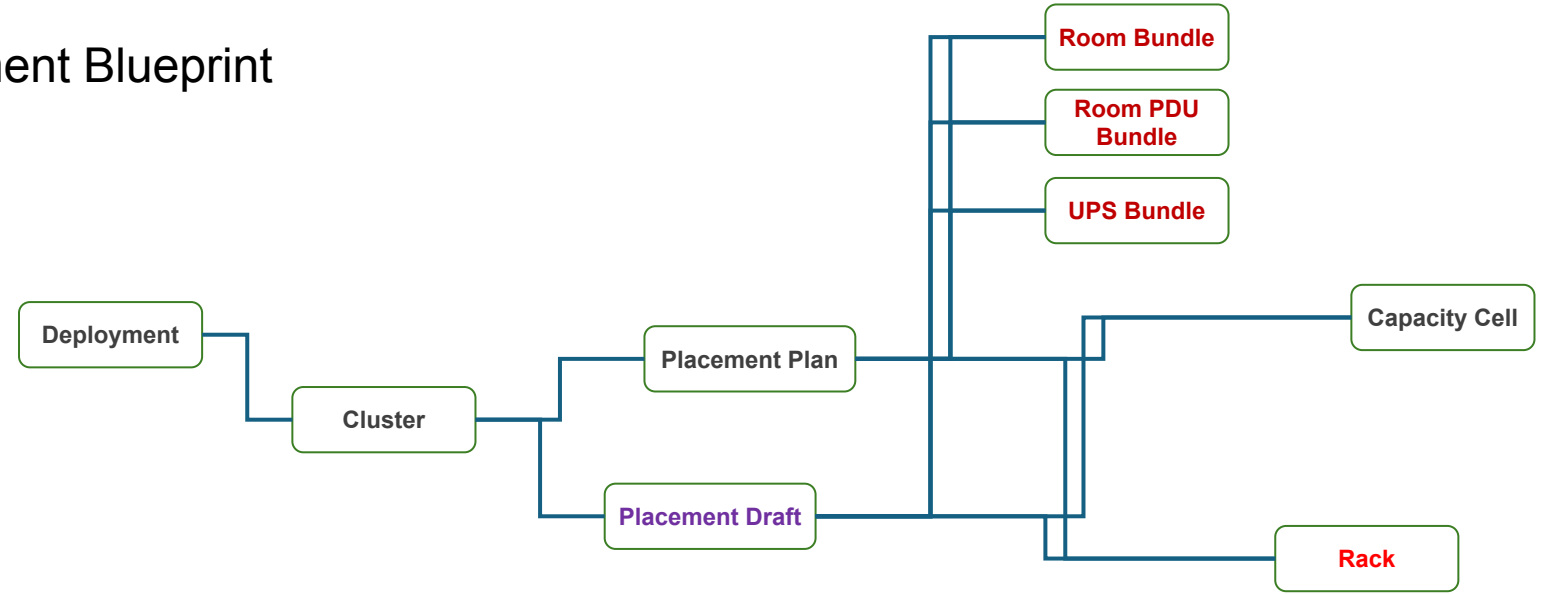
- Insert this segment to the electrical topology in the next Sprint

Crate synonym pairs:

- RPP – Busway Segment
- Branch Circuit – Busway Tap

Some data centers may use the RPP & Branch Circuit structure
Some data centers may use the Busway Segment & Busway Tap structure

Deployment Blueprint



- A cluster can have 0~1 **placement plan** and 0~many **placement drafts**
- A placement plan/ draft include **Capacity cells** and (sometimes) **"free-floating" racks.**

What is **"Free-Floating" Racks?**

In the Whitespace Blueprint, all racks belong to and belong to only one Capacity Cell.

When building Placement Plans, when the Capacity Cell is used, all the racks of the Cell will be associated with the Plan.

However, occasionally the planner may choose only to use individual racks of a Cell (this breaks the integrity of the Cell).
So when racks are directly associated the the Placement Plan, they are called "free-floating" racks.

Branch Circuit modification (future)

- The only first modification of the infrastructure we allow the user to do is to change the RPP to rack wiring.
- To better track it, we need to reflect the actual relationship by adding a Branch Circuit topology
  - RPP multiple branch circuits, each has a rating (e.g. 60A)
  - A number of branch circuits would be assigned to a row: Branch Circuit Row
  - Only at the load placement time, the branch circuits would be liked to the specific Rack: Branch Circuit Rack
  - In real deployment, branch circuit is wired to the Rack PDU. But for planning purposes, we can use Rack.
- During phase 3 (placement), we will have a default Branch Circuit Rack link. But will allow the user to change it if necessary.

Old

Reservations

All SBOs and individual racks can potentially be reserved.

- They are reserved BY Placement Plans and Placement Drafts.
- A rack or SBO can only be reserved if their status is “free”

Reserving any SBO changes the status of itself and all its sub-elements including racks to “reserved”

Reserving any SBO changes the status of all its **affected** super-structures and **affected** peers to “partially reserved”.

Reservation Action	Rack	CC	CBD	Composite CC	Composite CBD
Rack	Reserved	Partially reserved	Partially reserved	Partially reserved	Partially reserved
CC	Reserved	Reserved	Partially reserved	Partially reserved	Partially reserved
CBD	Reserved	Reserved	Reserved	Partially reserved	Partially reserved
Composite CC	Reserved	Reserved	Partially reserved (affected peers)	Reserved	Partially reserved
Composite CBD	Reserved	Reserved	Reserved	Partially reserved (affected peers)	Reserved

Reserving any rack change the rack status to reserved, and changes the status of all its affected super-structure to “partially-reserved”

Release of Reservation

All reserved SBOs and individual racks can be released from the Placement Plan or Draft.

When any SBO is released, all of its sub-elements, including racks, are released, status changes to “free”.

When any rack or SBO is released, re-compute reservation status for each of its super-structures or peers that can potentially be affected. If all racks are free, then status changes to “free”.

Validation: no conflict between SBO and racks. For any racks to be “free”, all of its racks must be free. There should not be a conflict.

Reservation Status (RST) are Scenario-Dependent

Just like Allocated Load, Placement Drafts are allowed to have their ”private” reservation status. Changing reservation status in the “private” PD scenario does not affect ”public” reservation status.

For each Rack and SBO, there is an RST_{public}, which is based on ALL the placement plans.

There are $RST_{PD,}$ one for each specific Placement Draft.

How Rack.OAKw is affected by reservation status

In APR Calculator, if a Composite SBO is partially reserved, it contributes 0 to the composite count.

If a Composite SBO is free or reserved, it contributes 1 to the composite count.

If a Base SBO is partially reserved, its base premium rate will be set to 0%, therefore the APR will also be 0%.

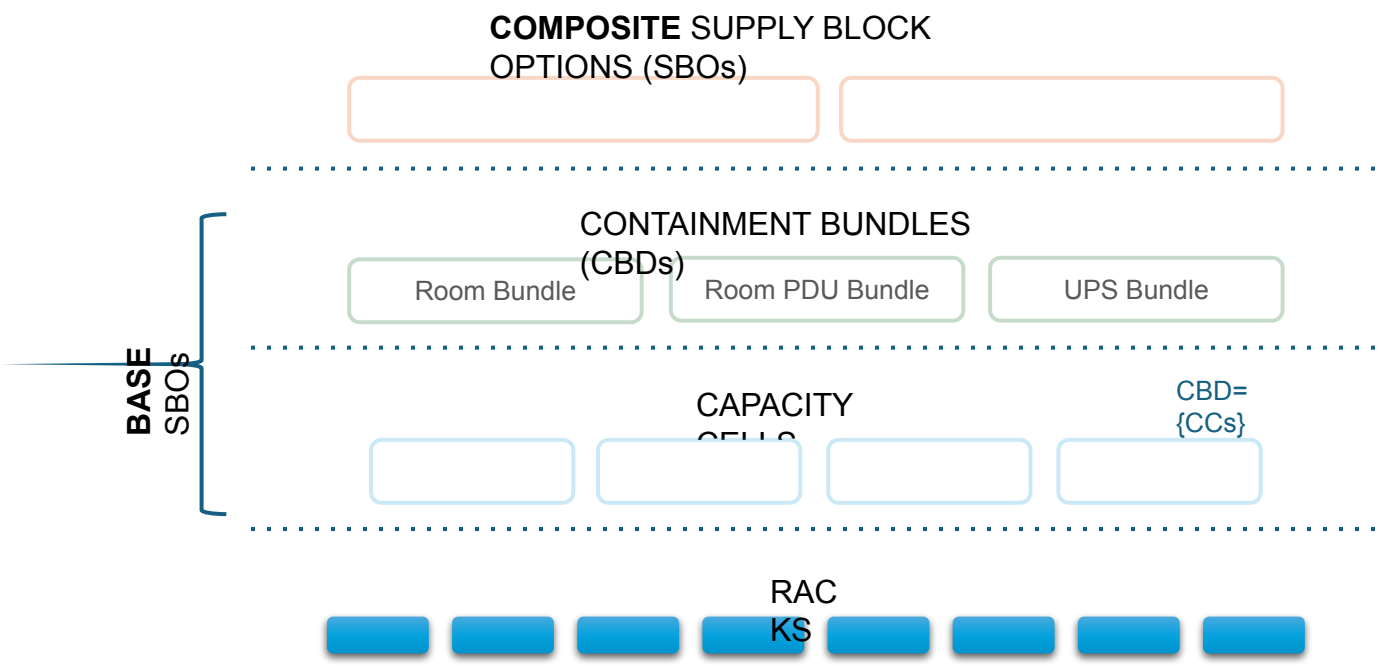
Since Reservation status are scenario based, Rack.OAKw is also scenario based.

IT Facility.OAKw $_{(whitespace)}$ = total of Rack.OAKw in the facility, assuming all SBOs are free

IT Facitly.OAKw Rate $_{(whitespace)}$ = IT Facility.OAKw $_{(whitespace)}$/ Facility_Deployable_ITCapacity

IT Facility.OAKw $_{(residual)}$ = total of Rack.OAKw in the facility, according to each PP or PD's scenario

Reservations and Releases



Racks in:	**Penalized for**	**Rewarded for**
Capacity Cell	**Cross-room entanglement**: a single CC spanning over more than 1 Room	**In-room modularity: m**ore than one CC within a single Room
Room Bundle	**Cross-room entanglement:** a single Room Bundle spanning over more than 1 Room	
Room PDU Bundle	**Cross-room entanglement:** a single Room PDU Bundle spanning over more than 1 Room	**In-room modularity: m**ore than one Room PDU Bundle within a single Room
UPS Bundle		**In-floor modularity: m**ore than one UPS Bundle within a single Floor

7. Supply Block Membership Premium

Containment Class	Compound Membership	Compound Membership Premium (CMP)
Composite Capacity Cell	1	CCC Base premium = 5%
	2	
	3	
	n	
Room Bundle or Composite Room Bundle	1	Room Bundle Base premium = 2%
	2	
	3	
	n	
Room PDU Bundle or Composite Room PDU Bundle	1	Room PDU Bundle Base premium = 3%
	2	
	3	
	n	
UPS Bundle or Composite UPS Bundle	1	UPS Bundle Base premium = 3%
	2	
	3	

Calculation Examples with the current topologies

Supply Blocks (Composites)	Components
CBDL-R-01	BDL-R1R2
	BDL-R3
CBDL-R-02	BDL-R1R2
	BDL-R4
CBDL-R-03	BDL-R1R2
	BDL-R5
CBDL-R-04	BDL-R3
	BDL-R4
CBDL-R-05	BDL-R3
	BDL-R5
CBDL-R-06	BDL-R4
	BDL-R5
CBDL-R-07	BDL-R1R2
	BDL-R3
	BDL-R4
CBDL-R-08	BDL-R1R2
	BDL-R3
	BDL-R5
CBDL-R-09	BDL-R1R2
	BDL-R4
	BDL-R5
CBDL-R-10	BDL-R3
	BDL-R4
	BDL-R5
CBDL-R-11	BDL-R1R2
	BDL-R3
	BDL-R4
	BDL-R5
CBDL-RPDU-01	BDL-RPDU-R1R2
	BDL-RPDU-R3R4
CBDL-RPDU-02	BDL-RPDU-R1R2
	BDL-RPDU-R5
CBDL-RPDU-03	BDL-RPDU-R3R4
	BDL-RPDU-R5
CBDL-RPDU-04	BDL-RPDU-R1R2
	BDL-RPDU-R3R4
	BDL-RPDU-R5
CBDL-UPS-01	BDL-UPS-01
	BDL-UPS-02
CCC-01	CC-R5E
	CC-R5W

Components	# of Supply Blocks including CBD	Base Premium	CMP	
BDL-R1R2	8	2%	**4.00%**	Each rack in BDL-R1R2 will receive a 4% premium
BDL-R3	8	2%	**4.00%**	Each rack in BDL-R3 will receive a 4% premium
BDL-R4	8	2%	**4.00%**	Each rack in BDL-R4 will receive a 4% premium
BDL-R5	8	2%	**4.00%**	Each rack in BDL-R5 will receive a 4% premium
BDL-RPDU-R1R2	4	3%	**6.00%**	Each rack in BDL-RPDU-R1R2 will receive a 6% premium
BDL-RPDU-R3R4	4	3%	**6.00%**	Each rack in BDL-RPDU-R3R4 will receive a 6% premium
BDL-RPDU-R5	4	3%	**6.00%**	Each rack in BDL-RPDU-R5 will receive a 6% premium
BDL-UPS-01	2	3%	**4.24%**	Each rack in BDL-UPS-01 will receive a 4.24% premium
BDL-UPS-02	2	3%	**4.24%**	Each rack in BDL-UPS-02 will receive a 4.24% premium
CC-R5E	1	5%	**5.00%**	Each rack in CC-R5E will receive a 5% premium
CC-R5W	1	5%	**5.00%**	Each rack in CC-R5W will receive a 5% premium

For **each rack**, add up all the CMP together, then +1 □ total premium

Whitepace_OAKw, Residual_OAKw, Draft Residual_OAKw

On the **Whitespace Blueprint topology:**

- WP_OAKw should be computed for each entire **IT Facility**
- If a Site has multiple IT Facilities, a Site WP_OAKw should be calculated by sampling adding WP_OAKw from multiple IT Facilities
- If Whitespace changes, WP_OAKw needs to be recomputed

On the Deployment Blueprint topology, calculate Residual OAKw (R_OAKw)

- Perform the same OAKw Evaluation process, but exclude the CC and/or Racks that have been committed to Placement Plans.
- If there are Placement Plan changes, R_OAKw needs to be recomputed

Also on the Deployment Blueprint topology, calculate Draft Residual OAKw (DR_OAKw):

DR_OAKw is specific to each deployment draft. Peform the same OAKw Evaluation process, but exclude the CC and/or Racks that

(1) Have been committed to Placement Plans

(2) Have been place in **THIS** draft

If a Deployment Draft changes, its DR_OAKw needs to be recomputed

Calculate Optionality Impact

Optionality Impact Definition: by placing certain CC and/or racks, how much does the OAKw value of the remaining facility change before and after?

Example 1

Assuming an IT facility consists of 1000 racks. 100 racks are being placed in a **Placement Plan or Draft**. The other 900 are still free.

Step 1. Run the OAKw Evaluator, assuming 1000 racks □ **Rack**.R_OAKw$_{before}$

(Since no other plans has taken any racks yet in this particular case, R_OAKw = WP_OAKw)

Step 2. Sum the Rack.R_OAKW$_{before}$ of only the 900 Racks □ IT_Facility_Available.R_OAKw$_{before}$

(*This is the IT Facility's R_OAKw$_{before}$, but only counting the 900 rack that will still be available after the placement plan/draft)*

Step 3. Run the OAKw Evaluator **again**, assuming only the remaining 900 racks □ **Rack**.R_OAKw$_{after}$

Step 4. Sum the Rack.R_OAKW$_{after}$ of the 900 Racks □ IT_Facility.R_OAKw$_{after}$

Example 2

Assuming an IT facility consists of 1000 racks. 100 has been placed in PP01. Now we have another placement: **PP02 or PD02 of 200 racks**

Step 1. Run the OAKw Evaluator, assuming 900 racks □ **Rack**.R_OAKw$_{before}$

Step 2. Sum the Rack.R_OAKW$_{before}$ of only the 700 Racks □ IT_Facility_Available.R_OAKw$_{before}$

(*This is the IT Facility's R_OAKw$_{before}$, but only counting the 700 rack that will still be available after PP02/PD02)*

Step 3. Run the OAKw Evaluator **again**, assuming only the remaining 700 racks □ **Rack**.R_OAKw$_{after}$

Step 4. Sum the Rack.R_OAKW$_{after}$ of the 700 Racks □ IT_Facility.R_OAKw$_{after}$

SJ-Meridian-C01 / SJ-Meridian-C01-PP01

kW Placed / Requested	Racks Placed / Requested
450kW / 450kW 100%	36 / 36 100%

Captured Capacity Value	Average CCV/ Rack	Remaining Portfolio Value (optionality adjusted)	Optionality Impact
$778,140	$43,230	$24,918,438	$ (31,126)

Total kW Requested	Total Racks Requested	Cooling Mode	Power Redundancy
450 kW	36	Air, RDHx	N

Rack Role Placement

Role	Target kW	Count	Placed	Cooling	Placement
Compute	15 kW	18	18	↔ Air, RDHx	CC-R1R2
Compute	10 kW	18	18	↔ Air, RDHx	CC-R1R2

Capacity Cells

CC-R1R2
920 kW guaranteed · 36/48 racks · 450 kW allocated

Power Load/ Power Capacity	Thermal Load/ Cooling Capacity
450/920 kW	450/940 kW

Power: RPP-R1-1, RPP-R1-2, RPP-R2-1, RPP-R2-2
Thermal: AIRZONE-R1, AIRZONE-R2, LL-R1, LL-R2

View Placement | Modify Placement | Run Auto-Placement | Commit Plan

Optionality Impact =IT_Facility_Available.R_OAKw$_{before}$ - IT_Facility.R_OAKw$_{after}$